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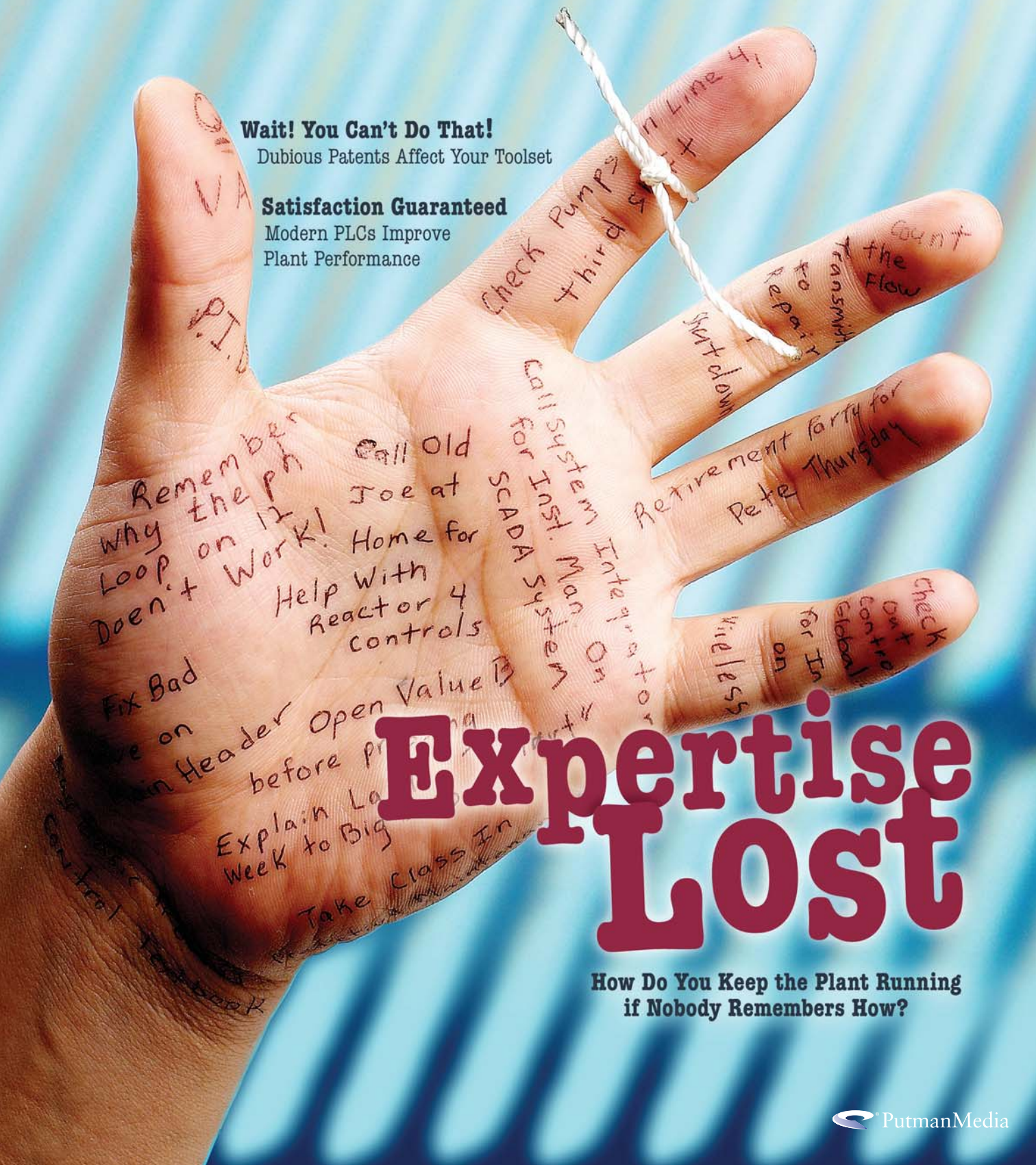
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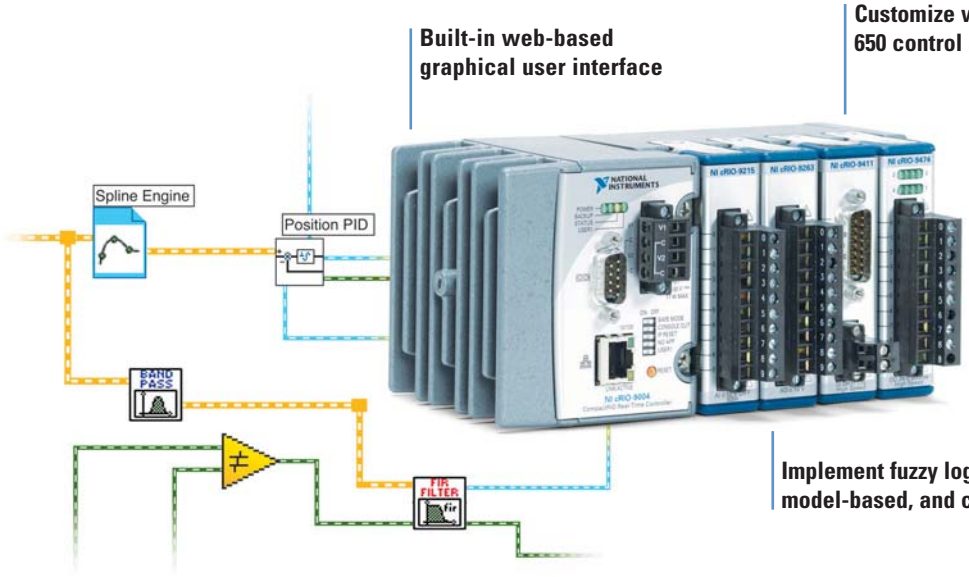
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Can Cheap Flowmeters Be Good?

A couple of months ago, in the company of Erik Rosaen of Flowmeters.com, I visited Fermilab (Batavia, Ill.). This Department of Energy research facility in the western suburbs of Chicago is one of the largest high-energy physics labs in the world. It is also the largest manufacturer of the world's single most expensive product.

In a hybrid continuous and batch process, Fermilab produces and stores antimatter particles called anti-protons. Its real-time process control system displays current values on the web site, as well as in its control room. As I write, Fermilab has a relatively small number (51.59 E10) of anti-protons in inventory, and the main beam is down.



Because of the highly specialized nature of the manufacturing process, Fermilab's instrumentation engineers originally built their own controllers. As you walk through the corridors, however, you can see their nearly "geological" progression from custom hardware and custom software to customized Allen-Bradley PLCs and PanelView displays, and finally to COTS (commercial, off-the-shelf) equipment from AutomationDirect.

The two critical control parameters are electric feed into the system, and water flow to cool the enormous set of magnets that cause the anti-protons to go where they are supposed to go. The reason Rosaen was there was that Fermilab is in the process of installing a large number of his inexpensive Coolpoint vortex-shedding cooling water flowmeters to measure this critical parameter, and he wanted to show them off to me.

The reason I was there was to ask the Fermilab engineers why they would use a "cheap" flowmeter to measure a parameter so absolutely critical to their cutting-edge, and very expensive, manufacturing process. Senior engineer David Peterson said, "It isn't the cost of the device but the durability that we're looking for. It doesn't have to be accurate, either, as long as it is very repeatable. If we lose cooling water, the process is shot, we have to dump the particles and start all over again. These flowmeters, because they have no moving parts, are very attractive from a durability point of view." In an effort to puncture Rosaen's bubble, Peterson added, "We've been through a lot of expensive flowmeters here, so ask us in a year how well these worked." You can see my full article on Fermilab and their flow measurement application on ControlGlobal.com.

Traditionally, one of those "expensive" flowmeter types has been the Coriolis mass flowmeter. Now several companies, including the biggest Coriolis vendor, Emerson Micro Motion itself (www.emersonprocess.com), have begun offering lower cost, and even very low-cost versions, in an effort to widen the application base for the instrument. AW Company (www.awcompany.com), for instance, has launched a Coriolis "price war" by dropping the price of select sizes of mass flowmeters to below the typical pricing of magnetic and other volumetric flowmeters.

Racine Federated (www.racinefederated.com) vice president Tom Nelson told me, "The movement is toward lower cost and improved performance, all at the same time."

I believe that Nelson is right, and these examples seem to reinforce my belief. Every time a vendor reduces the price of a process instrument, or introduces a lower overhead cost control system, it increases, sometimes tremendously, the number of processes that are now "worth measuring." As this happens, new opportunities are created for cost avoidance, increased productivity, and even profit. This level of increased competition benefits the end user and the integrator greatly. And the vendors can make it up in volume.



***Walt Boyes,* Editor in Chief**
wboyes@putman.net

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Dale Peterson's Security Column

Columnist Dale Peterson, from Digital Bond.

And if the Net Goes Boom?

SCADA Security is more than preventing hacking. What happens if the grid goes down? Can your system stay up? CONTROL's Walt Boyes reports.

EXCLUSIVE TO THE DFI:

Machinery Health Comes to Fieldbus

Walt Boyes comments on CMMS, asset management, and Emerson's new addition to PlantWeb and Fieldbus.



On the Bus

New column by Ian Verhappen, Chair of the Foundation Fieldbus User Group, and vice president of Standards for ISA.

EXCLUSIVE TO CONTROLGLOBAL.COM

Making the World's Most Expensive Product

At Fermilab, in Suburban Chicago, they make the world's most expensive product: Anti-matter. Yet, it is a production facility just like the Oberweis Dairy plant that makes ice cream about a mile away. CONTROL takes a look at how an anti-matter factory works, and why flowmeters are critical to the operation of the plant.



Blog This!



Take a look at *Sound Off*, CONTROL's editors' blog. Want to know what's really going on? ControlGlobal.com's entry into the blogosphere is now up and running. Here's where CONTROL's editors get to post their ramblings and you get to respond. Read what's going on and feel free to respond to any of the subjects being discussed. We'll post interesting responses.

Ask the Experts

CONTROL's Béla Lipták, along with his cadre of process automation experts, moderates "Ask the Expert" and provides answers to readers' knotty control automation problems and career issues.

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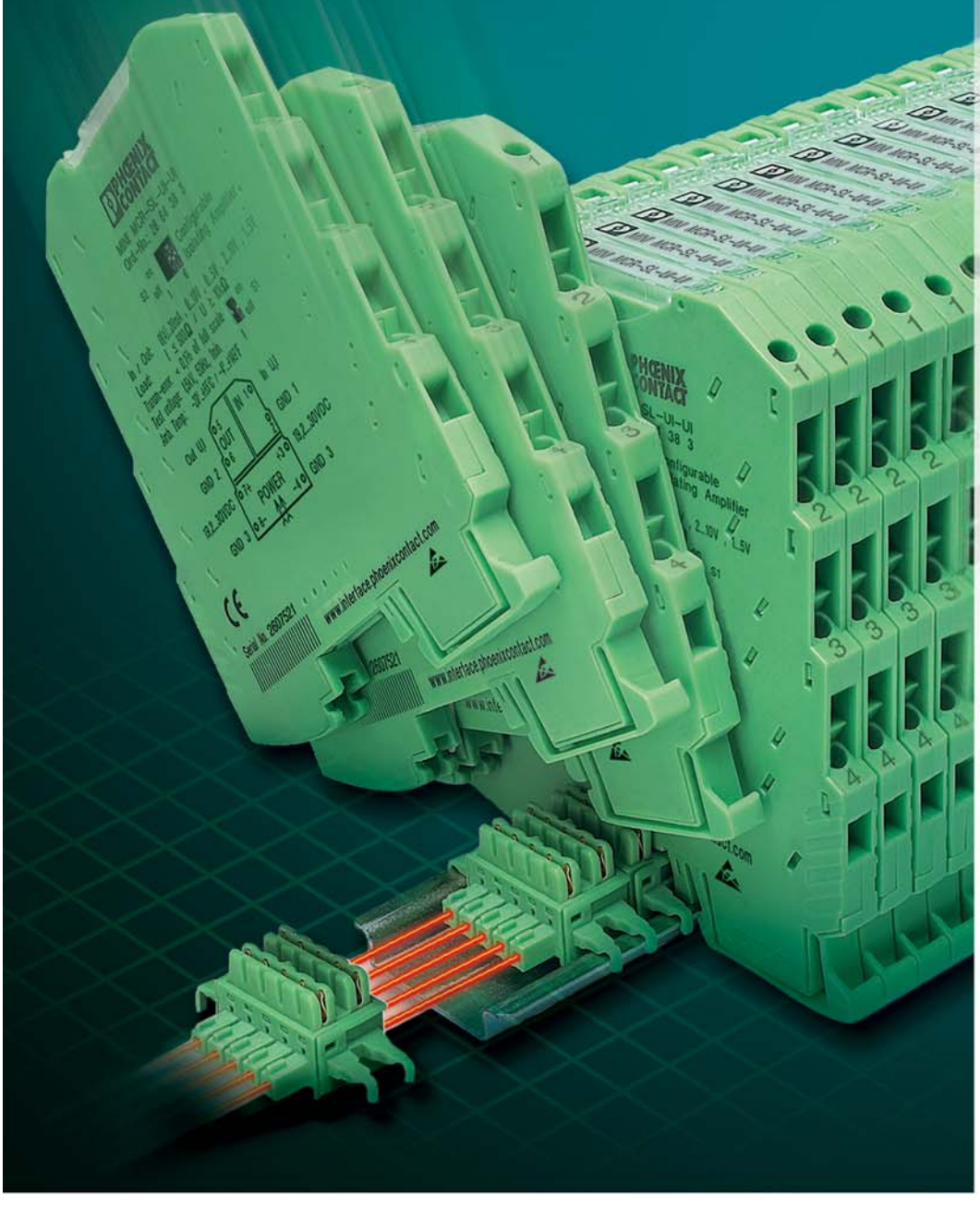
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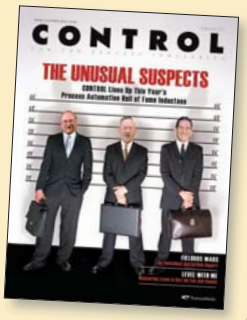
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Bloggin' the Drag

I read in your blog your comments about "drag factors" and the ARC Forum ("Sound Off," 2/2/2005, www.controlglobal.com). I agree completely! Bravo!



I am an engineer with 30+ years in the field, and run into drag factors every day. One of the worst is that the middle managers who approve/recommend projects have no idea what SCADA/HMI can do for them. Most are products of business school and don't know what I/O, SCADA, PLC and other acronyms mean.

To their thinking, the machine is making reasonably good product right now, so what benefits will a costly modification get them? I can tell them: better quality, more uptime, better machine utilization, real-time management information, etc. When I suggest this all I get in return are blank stares. Most have no experience with a system that really gives a good return, so how does one go about educating these middle managers to show them that, one, the modification can really deliver; and two, that it will be worth the time and expense of implementing it? The manufacturers of SCADA and even HMI systems need to sell to these middle managers as well as the engineers who will choose which product. It does little good to just sell to the engineer, when he/she can not get the project funding approved.

Richard D. Carey, PE
Sr. Process Engineer
Michelin North America

Brand Is Reputation

Your editorial on brands ("In Our View," Feb., p11) was superb; it hit me at the right time. I'm just starting out, in a way. I believe that what is important to a new business is to always be aware that a reputation is a business' best advertisement and should always be nurtured and protected. I'm not sure if brand and reputation are the same thing. You can have a reputation without having a brand, but can you have a brand without having a reputation? (probably not). Thanks for the insight.

Robert C. Rosenbaum, PE
American Canyon, Calif.

Chinese Currency Shenanigans

Charles Schumer (D-NY) and Lindsey Graham (R-SC) have drafted legislation to impose an import tariff on Republic of China imports. China has pegged its currency to the U.S. dollar since 1994, despite an exponential increase in the U.S. trade deficit with the country. The peg has kept Chinese imports artificially cheaper than U.S. goods and has cost thousands of U.S. manufacturing jobs. Experts say that the Chinese currency is undervalued by 40%. It seems that a certain senator from Iowa, Charles Grassley, the ranking Republican member of the senate finance committee, has prevented this legislation from leaving committee. The sponsors of this legislation believe they have enough votes to pass it in the Senate.

I have written to Sen. Grassley's office, urging him to consider supporting this legislation.

I am a CONTROL magazine subscriber and have enjoyed your no-nonsense articles on the plight of U.S. manufacturing, and thought you and your readership would be interested in this matter.

Al Rogers, PE
Newark, Del.

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Jimmy Neutron Meets the Woz

James Isaac Neutron is the hero of a popular animated television series on Nickelodeon. A feature film titled "Jimmy Neutron: Boy Genius" was nominated for an Oscar. Jimmy will also be an attraction at Universal Studios in Florida, replacing the Hanna-Barbara ride. And I am a fan.

I first met Jimmy cruising the satellite TV while recovering from knee surgery. Whizzing by the Nickelodeon channel, I saw Jimmy in his pre-puberty classroom. The classroom blackboard had on it some math and vocabulary lessons. These blackboard items were sophisticated and correctly presented—most surprising. "This can't be a kids cartoon," thought I. I looked, hesitated and was hooked.



"What do you make here?" cynically asked the visitor. "Eighteen percent," was the answer.

What has this to do with this magazine? We are control geeks, not kids. Oh no? Jimmy tries to control his town, Retroville, with his high-tech genius. His purpose in life is to avoid simplicity and pile on the gadgets.

In one of the TV spots Jimmy's mother wants him to hang up his pants. Jimmy spends an inordinate amount of time applying nanotechnology and chaos theory to make the pants hang themselves. The now-intelligent pants start a revolution with the help of the local clothing store. So Jimmy must invent a super static cling gun to put down the revolution. After all of this adventure, Jimmy's friend, Carl, suggests that maybe it would have been simpler to hang up the pants himself like his Mom asked him to.

The Jimmy Complex

Engineers all over the world have a Jimmy complex. We have an outdoor spa at my wife's house, where I live. We use this spa all year round, including during the New Hampshire winter. Our winters have lots of snow. Last year's snow load on the spa broke the insulating cover.

Me, being the engineer, gleefully launched into a redesign of the "bad" cover. The materials budget alone now exceeded the price of a new cover. My long-suffering bride suggested a PVC pipe internal to the spa as a mid-support for the cover. It took me a while to suppress my testosterone to accept her solution. But, alas, I could not leave it alone. I ended up with an adjustable PVC pipe with feet on both ends. Feature creep is everywhere. I was Jimmy Neutron with the pants.

Why did I do this? I can't help it. My engineer DNA operates as an independent embedded software application, and I can't shut it down. I have the feeling that my DNA came out of Redmond, Washington.

Taking the Cure

Last fall, I took the cure. I attended an emerging technology conference at MIT. I even slept at the MIT hotel—an experience all by itself. The conference hosted speakers concerned about the impact of new technology upon the business, political and economic issues. The speakers represented large companies, technologists and geeks. To name-drop but a few of the speakers: Managers from GM, IBM, Xerox, 3Com and Kurzweil were the opening keynoters.

Most of the suits wanted to purchase continued corporate success with the currency of innovation. They focused upon the need for discipline, budgets and "faster, better, cheaper" systems and products; not truly disruptive thinking.

What the speakers did suggest was that the "age of invention" is coming. A renaissance in technology innovation will be the thrust for the next several decades. Will we, the control geeks, be ready? What must we do to stay out of the sandbox and participate in the cocktail party of true innovation? An answer was suggested at the MIT foray.

Steve Wozniak (the Woz), who helped shape computers by designing major portions of the Apple Computer, had some suggestions:

Never design anything you can't throw away. The key to innovation is being lazy. Fewer parts means less cost. Cost reduction is a design function, not an efficiency issue. Solving problems takes more than horse power. Computers (and control systems) are defined by software. Computers are telescopes, microscopes and time machines "plus God."

The Woz suggests that the control engineer is more than a lackey of the MBA or CFO. The idea of production and manufacturing costs being the efficient replication of poor designs is flawed. We need to think and innovate from lust to dust. Design needs to be woven into control considerations.



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I am reminded of early PLC manufacturing. We were very efficient, but had low profits. We were, as IBM puts it, in "commodity hell." Efficient and flawless, yes, but the protein of profit was missing.

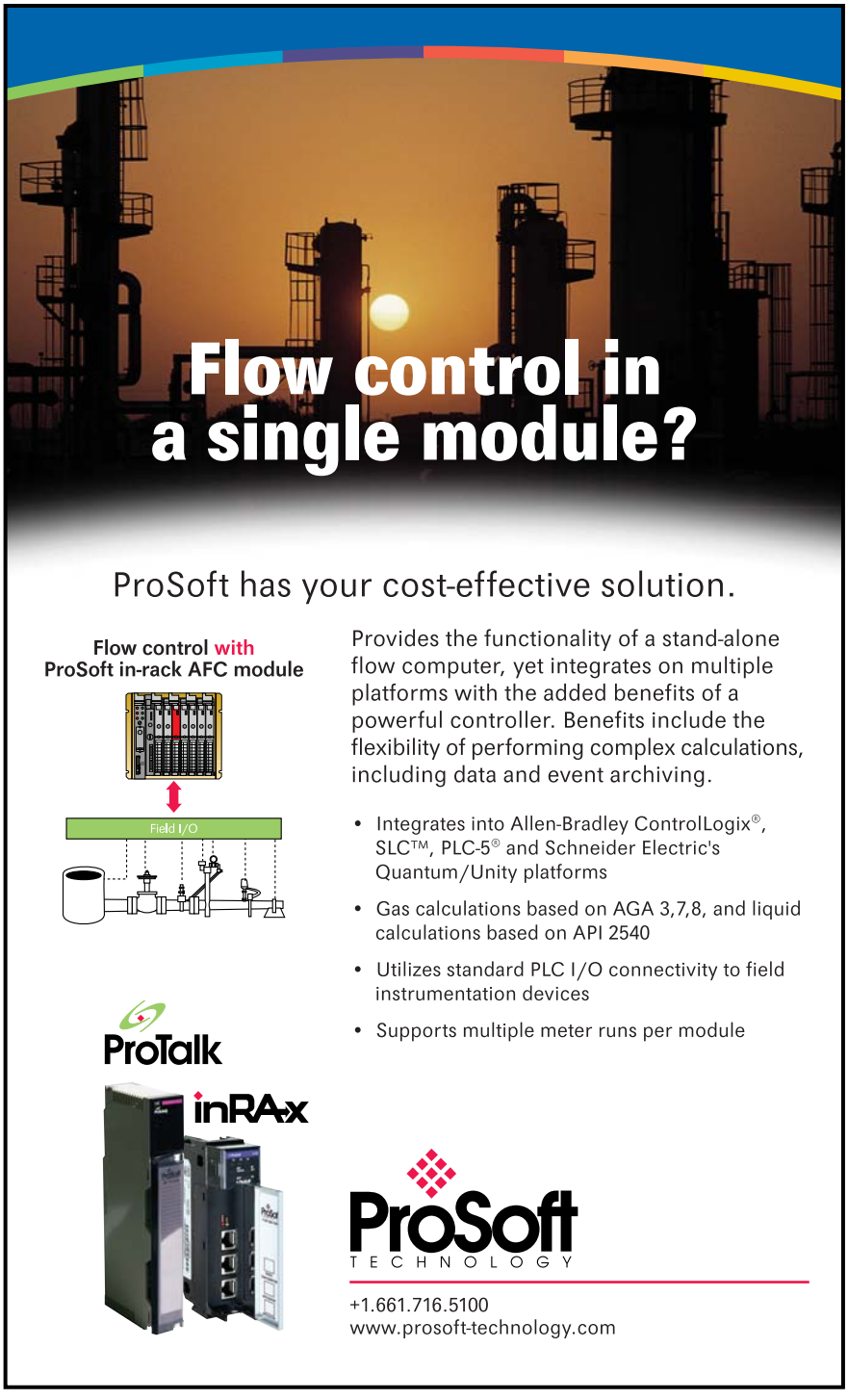
Later, a new company revisited a controller design used for buildings and substantially removed parts and basic costs compared to the competition. Visitors would stand in awe watching the PLC manufacturing floor, and sneer when they visited the new company at the building control assembly.

The new company's assembly floor was a mess—with parts and other things strewn about—it had the look of a diner kitchen. "What do you make here?" cynically asked the visitor. "Eighteen percent," was the answer. You can overcome the lack of apparent inefficiency with good high value design. We achieved one-third the cost with three times the profit.

Old as Woz

The lesson for us is to be as old as the Woz (he says he's eleven now) and think the whole problem through. It is tough for us to sell the halls of management, but we need to do this. We need to understand the real needs of the enterprise and communicate a strong, well thought-out business case for simple designs leading to simple factories.

Don't let today overwhelm the future, KISS.



Commonly credited with creating both the modern PLC and building automation control system, Dick Morley quit working so he could wear t-shirts and ride his Hawg, or go skiing on Thursdays, depending on how he tells the story. He lives in Shirley's house and works in a barn in New Hampshire. Contact him at morley@barn.org.

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Shake, Rattle and Roll 'Em Out the Door

RFID in the Process Industries?

By Walt Boyes, Editor in Chief

Joe Owen, Sujeet Chand, Andreas Somogyi, Vivat Bapat and Matt Bauer, Rockwell Automation's steering team for their RFID initiatives, provided a press briefing to CONTROL on the state-of-the-art in RFID for the process industries on February 28.

Rockwell (www.rockwellautomation.com) got out of the RFID sensor and reader business in 1999 after 15 years of producing chips, readers and associated peripherals. But their customers have kept asking them to continue, and now, it appears, Rockwell has been being asked by their own customers to help them implement the Wal-Mart and Department of Defense RFID initiatives.



Alien and other companies have a good shot at making money selling the technology but Rockwell is finding that creating real change in business processes and providing real solutions to current business problems will provide the long-term gain and the long-term business model for RFID.

According to Joe Owen, director of Strategic Marketing (read "Rockwell's Corporate Fireman"), Rockwell has decided to approach it as a consulting and systems integration type of business. They already knew, he said, a lot of the *don'ts*, which are often more important than the *do's*, by virtue of their former tenure in the business. This allowed them to jump in at a fairly high level, Owen opined.

According to Vivat Bapat, Rockwell's director of Supply Chain Solutions, the importance of RFID is not the technology of radio frequency tagging. It is what the easy availability of automatic unit identification can do when it is backed up throughout the enterprise.

This might all be vendor hype, as so many initiatives in the past have been. Last time it was RTPI and the ability to have a completely visible enterprise. The time before that … well you get the idea.

"The major change is not RFID, but the fact that we can Internet-enable RFID technology," said Sujeet Chand, Rockwell's chief technology officer. "Using RFID as a tag, as a 'URL' if you will, enables us to begin to hang lots of information on the tag in linked databases all over the world." Explained Chand, "It's like sticking cars onto a train. Every time the chip goes through a reader, another car is added to the train."

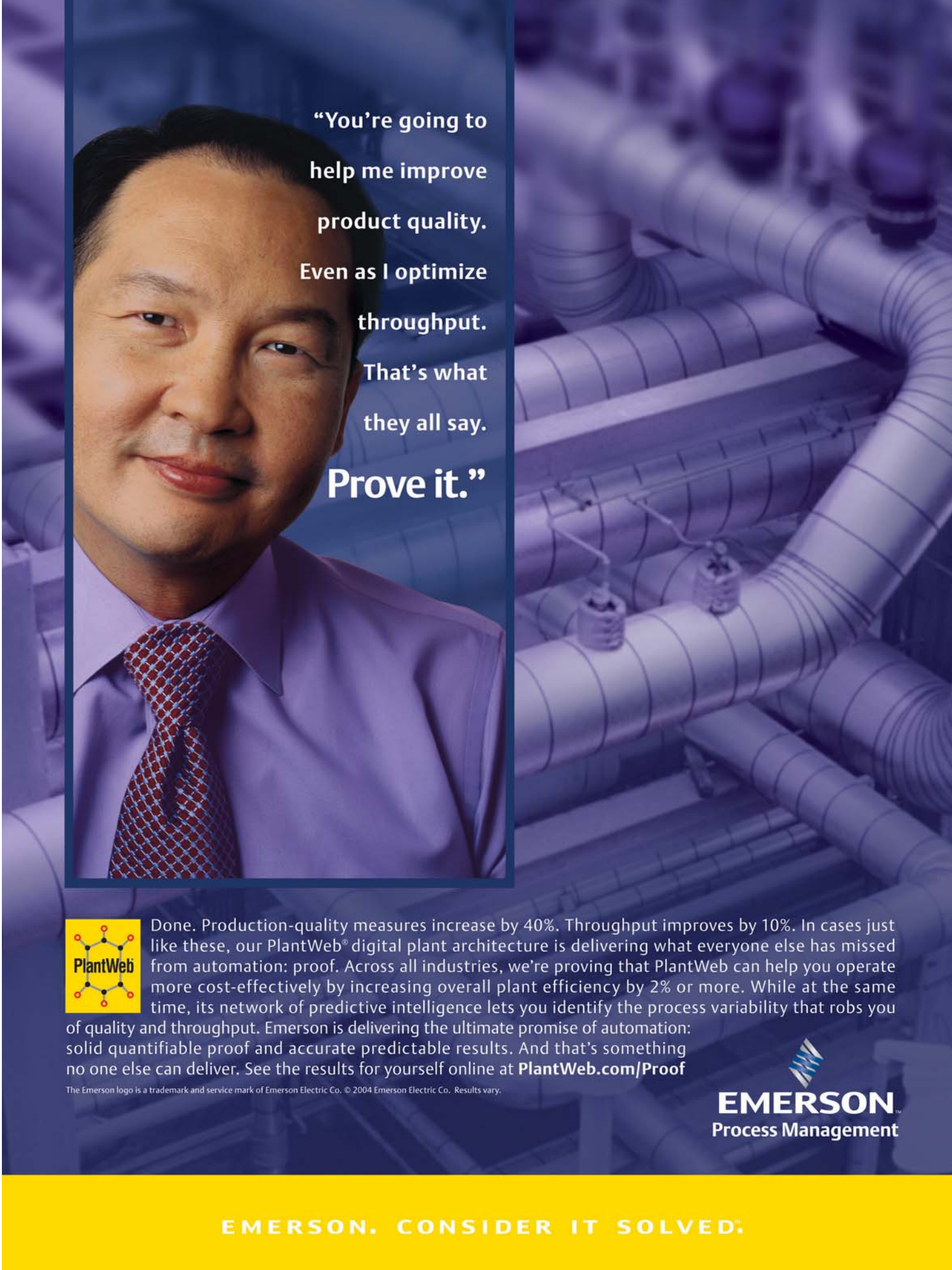
The implications of this are staggering. Is RFID the panacea, the next great thing that will save manufacturing? NO. Not only "No," but resoundingly so, according to Andreas Somogyi, Rockwell business consultant.

"Yet it is already heading down the wrong (technology) path as people spend more time talking about how the RFID chip can be used, rather than the business issues it will resolve," said a report by AT Kearney Inc. Joe Owen used this quote at the briefing to illustrate why Rockwell isn't getting back into the sensors and readers business.

Sure, Alien and the other companies are going to make a lot of money, hopefully, but Rockwell is quite sensibly correct, as Matt Bauer, Rockwell's director of Information Solutions Marketing pointed out, it is the real changes in business processes and the solutions to business problems that currently exist that will provide the long-term gain and the long-term business model for RFID. Rockwell, therefore, is sticking to consulting and systems integration.

The value, according to Bauer and Bapat, will come from anti-counterfeiting, track-and-trace, advances in quality monitoring, and the integration of RFID throughout the enterprise.

Alfonso Gutierrez, from the University of Wisconsin eBusiness Consortium, also attended the briefing. He commented, "RFID, and other AIDC technologies will have the effect of moving manufacturing from a transaction model to a placement model. That is, the fact that a



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package is located in a specific location, such as receiving, or factory service, will cause decisions to be made, based solely on where the package is. This is the most significant advance in manufacturing theory since we learned to debit all the components of the BOM from inventory automatically when a finished product shipped."

The fact that it is becoming obvious there is little or no ROI coming from the "slap and ship" model that is intended to meet Wal-Mart's directive isn't terribly exciting news. Pretty much everybody looking at RFID could tell that was coming. Joe Owen pointed out that the real ROI will come from people who say, "Well, we have to do this to satisfy Wal-Mart, or the DOD, or General Motors ... what else can we do with this technology?"

Process Control for Rotating Machinery

Emerson announced the ability to integrate AMS data into PlantWeb via a Foundation Fieldbus-enabled field device, the CSI 9210 Machinery Health Monitor that not only monitors rotating machinery for vibration, temperature, etc., but runs expert system software (of the proprietary type, of course) to determine and forward to the control system Machinery Health status, rather than raw data from sensors. The price point for a single machinery train will be between $5,000–$7,000 depending on configuration. At National Manufacturing Week in Chicago, GE Energy (formerly Bently-Nevada) announced a system that will do likewise, for about the same price tag, but without the simplicity of a fieldbus transmitter doing the work. Invensys proclaimed the same features for ArchestrA. Other manufacturers will clearly follow suit.



The real issue for expert system-based rotating machinery health monitors (like the Emerson CSI unit shown here) is that their price is over the line for capital purchase items. When Emerson and other vendors explore marrying these types of devices with wireless mesh networks, and provide them under $1,000 per point, these monitors will become much more interesting and you'll likely see more guys like this one installing the units on plant equipment.

For people who have fieldbus systems throughout their facilities, or who have industrial Ethernet throughout (system works with IE and Modbus, too) these are a great solution for absolutely essential rotating machinery trains, especially the predictive maintenance software embedded in the new product.

The real issue that GE, Emerson, et al., will have to overcome, though, is that each of these systems is over the line for capital purchase items in almost every brownfield plant in North America and Europe. It may not be possible to create serious justifications to be presented to management on the ROI possible by instrumenting just one machinery train.





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When Emerson and other vendors explore marrying these types of devices with wireless mesh networks, and provide them at a price point under $1,000 per point, these monitors will become much more interesting. This would make it a maintenance operating budget decision, not a capital purchase item. CONTROL predicts that long lines would form to buy the technology then.



Plant Automation Becoming Part of the Business System?

It has been a commonplace in process automation that the biggest problem with enterprise IT staff is that they don't understand the plant. Based on what SAP and Microsoft were saying at National Manufacturing Week, this is likely to change.

Jay Krueger, SAP business development manager for Machinery, Engineering and Construction, noted that he, himself, is a graduate of the process automation school of hard knocks, and understands the plant floor. William Bowers, vice president, Solutions Marketing – Manufacturing, and Konstantin Penz, vice president, Strategy, both affirmed SAP's new interest in, and overwhelming expertise in, process automation as one of the business systems in the enterprise. Charles Johnson, along with SAP alumnus Celestine S. Vettical, from Microsoft, and several of the major automation suppliers, went out of their way to show that they clearly understand how to tie the plant floor systems to the MES layer, and the MES layer to the enterprise.

CONTROL awaits developments.

The Drive for Drives

Two giants made new product family announcements at National Manufacturing Week, and CONTROL found the way the announcements were made of interest.

Siemens Energy and Automation, Inc. produced president and CEO Aubert Martin to underscore the importance of the introduction of their new product line of Sinamic drives. Held at a posh executive briefing room a bus ride away from the show, editors

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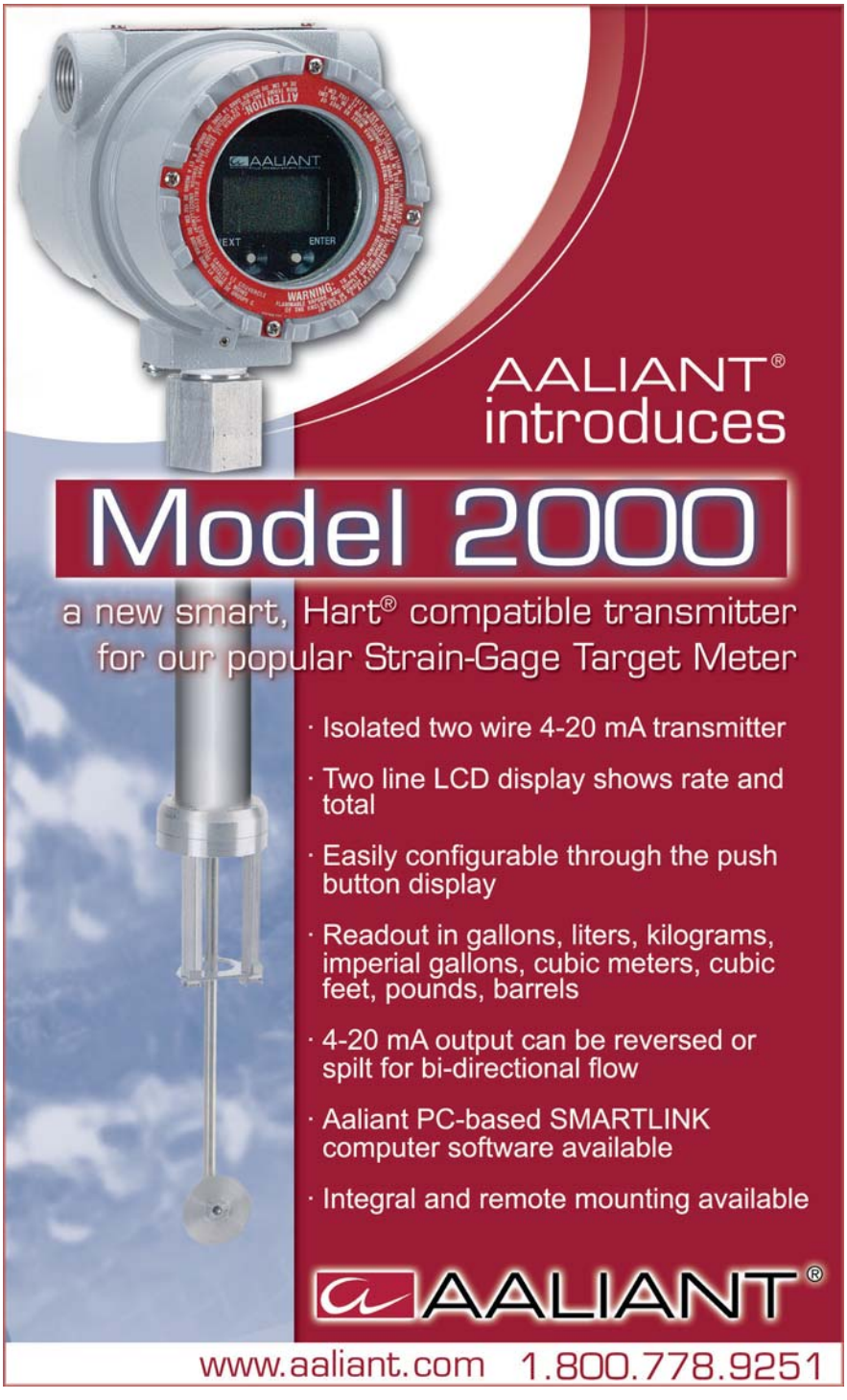
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were presented with a detailed seminar on marketing of new drive technology.

Baldor trotted out chairman and CEO John McFarland at an after-hours party to introduce their own new platform, the H2 drive product line.



Siemens made it clear that they were aiming the process and factory automation part of the new Sinamic product line at OEM and system integrator customers. It remains to be seen how the integrators will take to it. Machine builders who already use Siemens PLCs will find much to like in the new, backplane integrated drive designs.

Baldor spent money on both product function and on look and feel. Clearly the winner in the product beauty contest, Baldor's deep black drives with yellow logo and integrated yellow keypad and programmer, will be clearly visible to operators throughout the plant. Baldor, who already supports their OEM and integrator partners, as an associate member of CSIA, hopes to maintain their lead in drives for process and factory automation in the USA.

Siemens made the telling point that since they make one of everything (as does Emerson, and ABB, just for completeness' sake) it makes wonderful sense to standardize on Siemens for the One Big Automation Vendor in your plant, and Profibus/Profinet as the One Big Network inside it.

Controls Conference Set to Go

SPS Electric Automation America show organizers, announced the first North American program in more than 20 years to focus specifically on electric automation with a spotlight on controls, electric drives, sensors, software and motors. The Conference is to be held May 23–26, 2005, at the Donald E. Stephens Convention Center in Rosemont (Chicago), Ill. For more information visit the conference's web site at www.ea-america.com.

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—Edward R. Sederlund, Process Automation Product Manager, The Dow Chemical Company



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Uncle Sam Gets a D+ on Cyber Security

The department of homeland security led a list of seven agencies that received flunking grades for their cyber security efforts in 2004, with the federal government at large earning a "D-plus" from the House Government Reform Committee's annual cyber security report card.

Agencies that received failing marks include the departments of Agriculture, Commerce, Energy, Health and Human Services, Housing and Urban Development, and Veterans Affairs. A grade of "D" was awarded to the departments of Defense and Treasury, as well as the National Aeronautics and Space Administration and the Small Business Administration.

The grades were based on internal assessments by the agencies and evaluations by the White House Office of Management and Budget. Agencies were graded on how well they met the requirements set out in the Federal Information Security Management Act (FISMA). The law requires agencies to meet a wide variety of computer security standards, ranging from operational details—such as ensuring proper password management by workers and restricting employee access to sensitive networks and documents—to creating procedures for reporting security problems. This year's overall grade of "D-plus" was up slightly from last year's "D" and the "F" grade Uncle Sam earned on his report card in 2002.

Committee Chairman Tom Davis (R-Va.) said he was encouraged by the fact that 10 agencies improved their scores over 2003, increasing the overall government-wide grade by 2.5 points this year. But he chided agencies for not moving fast enough. "I hope it won't take some kind of major cyber attack to wake everybody up," Davis said.

Eight agencies earned lower grades for 2004. The departments of Commerce and Veterans Affairs saw their marks drop from a "C" two years ago to an "F" in 2004. One explanation for the lower grades, according to Dennis Heretick, the chief information security officer for the Justice Department, is that agencies were required to meet new standards last year that were not evaluated in past report cards, such as determining how frequently agencies applied software patches to fix known computer security flaws.



Expert System Helps Dow Save Energy

By Steve Kuehn, Managing Editor

Continuous use of one of the first closed-loop optimizer and expert system combinations in the petrochemical industry at Dow Chemical's Seadrift (Texas) Union-Carbide Corp. petrochemicals plant is saving $1.25 the company million in energy costs per year. Dow realized the savings after implementing Gensym Corp.'s G2 object-oriented expert system software platform. "The company tried open-loop optimizers but wasn't satisfied with savings it was generating because busy operators often didn't have time to fully implement the optimizer's suggestions," said Lowell Hawkinson, Gensym's chairman.

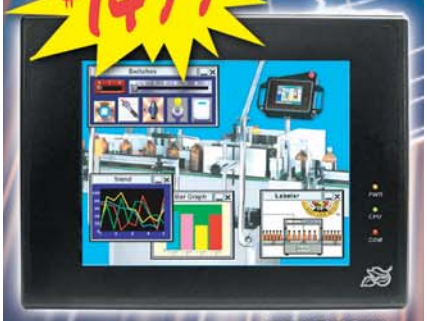


The optimizer Dow implemented recently quantitatively evaluates continually fluctuating electricity prices and the steam demands to determine the appropriate level to run the heat recovery steam generator on the company's cogen unit shown here.

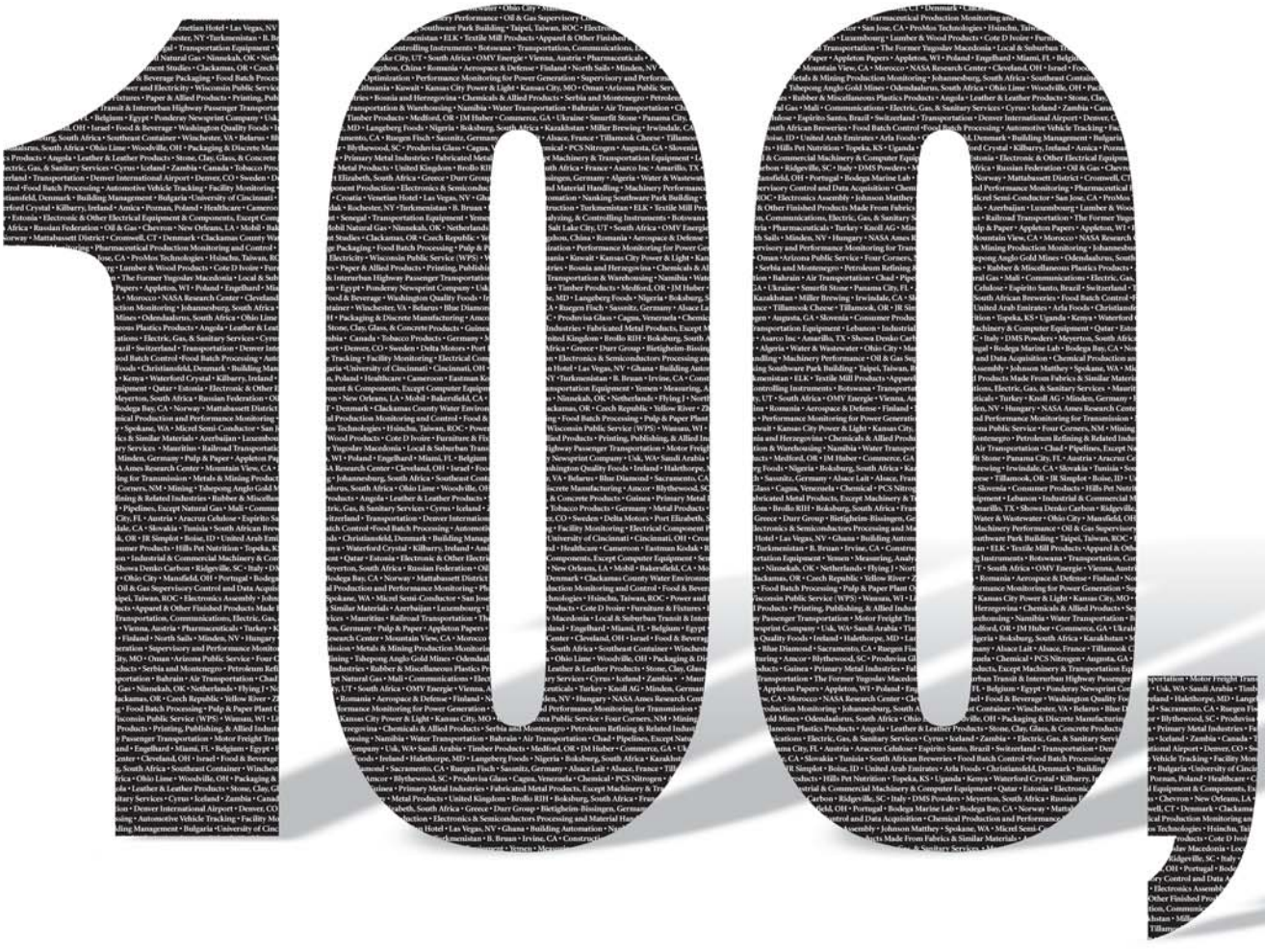
According to James Sturnfield, Union-Carbide senior specialist, savings have been achieved in the operation of Heat Recovery Steam Generators (HRSG) that create steam from the hot exhaust of a gas turbine for plant heating or to generate electricity via steam turbines. In manual control mode operators tend to run at a level that is comfortable to handle the swings in steam demand. This results in the steam turbines swinging their electrical loads as their steam loads change inversely with the plant steam changes. The optimizer, on the other hand, quantitatively evaluates continually fluctuating electricity prices and the steam demands to determine the appropriate level to run the HRSG and best utilize the steam generators. Then, with the aid of the real-time expert system, the optimizer automatically adjusts the load to achieve the best savings. The optimizer and expert system change operating setpoints every five minutes, if needed, while operators typically only had time to make changes once or twice per day.

Last year the Seadrift plant ran in closed-loop mode 98% of the time, which is remarkable for a plant of its complexity. The savings that have been achieved there are just the beginning since Dow believes it can more than double them by utilizing the optimization for more decisions and then multiply those savings across other plants.

Gensym, which recently launched the next-generation of its G2 platform, version 8.0, says the software delivers new reasoning and integration capabilities.

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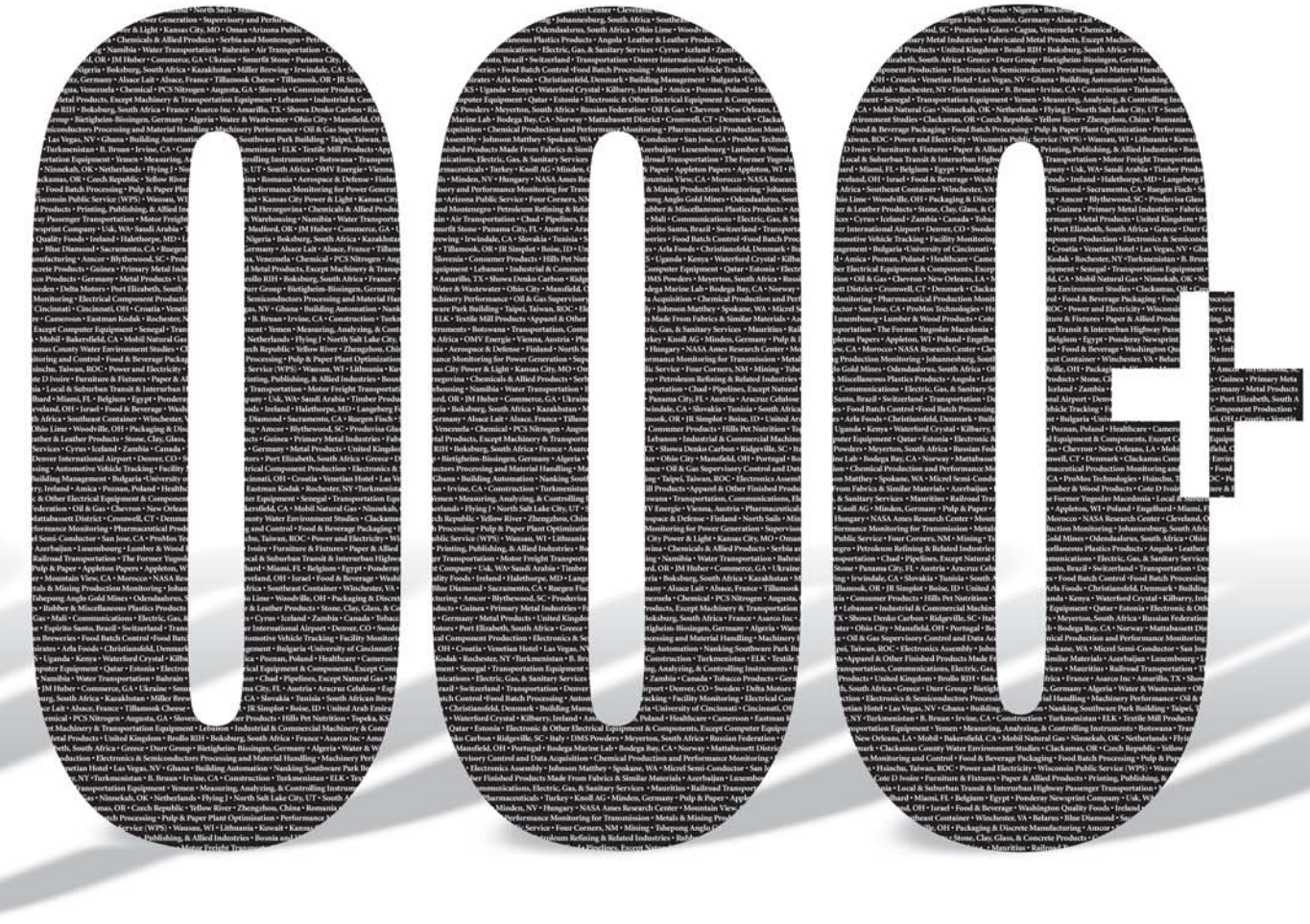
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King Coal Is Back!

Want to make $100,000 a year? Become a coal miner. According to National Public Radio (NPR), coal miner jobs are going begging. Many mines have been shut down for 50 years, but the increasing demand for coal is opening them up again. Problem is, there aren't any coal miners left, so mines are paying top wages. According to NPR, anyone willing to work overtime can make $100,000 per year in the mines.

For the same reason, plenty of jobs will also exist for process control engineers. Going into coal processing control may be the kind of career opportunity that only comes along once in a lifetime.



"Going into coal processing control may be the kind of career opportunity that only comes along once in a lifetime."

Thomas Sjudajski, director, Invensys Global Power Marketing (www.invensys.com) explains. "During the '90s, most new electric generation capacity was via gas-fired, combined-cycle plants. Many engineers who have spent the last 10 years in power generation have never worked on a coal plant. Consequently, the 'tribal' knowledge has not been passed on to many of the engineers who will be working on this new generation of plants."

Al Novak, business development manager of new coal-fired generation for Emerson Process Management (www.emersonprocess.com), notes: "Over the last 25 years or so, there has been little to no new coal-fired plant construction. This coal comeback presents some challenges for plant designers, engineers and operators."

Loss of process knowledge is a problem everywhere, as we discuss in this month's cover story, but it looks to be even more severe in coal. "One notable challenge is 'brain drain' caused by the retirement of those who have experience designing coal plants," says Novak. Another consideration: "The majority of proposed plants will utilize new technologies that did not exist when most current plants were originally commissioned."

Roger A. Leimbach, director of Marketing & Applications at Metso Automation (www.metsoautomation.com), agrees. "Coal-fired supercritical steam generators require unique control strategies, unlike other boilers, and process control engineers will be called upon to implement them," he says.

In a nutshell, King Coal is back with a vengeance, and there are only a few control engineers who know how to deal with it. You could be one of them because, even if you don't know coal control yet, you know all the other advanced control techniques that will be needed, such as asset management. "A significant area of activity is fleet-wide asset management for power generation companies, says Jil Shingledecker, project manager, Siemens Westinghouse Power (www.siemens.com) "These companies are optimizing their generating assets from fleet and plant standpoints."

And there should be plenty of jobs. "In North America there are now 96 proposed new coal-fired plants, representing $69 billion in investment," says Novak. That doesn't include all the old plants being modernized, or the explosive growth in Australia and China, which are building coal-fired plants in record numbers. "The power generation controls market in China is estimated to total $120 million annually, with projected double-digit growth," notes Novak.

That's a lot of control systems being installed, and they will all need your kind of experience. Fortunately, it looks like many of the control systems being installed are the same as in process control, so you should be right at home. For example, Foxboro is installing I/A Series controls in coal plants in New Mexico, Wisconsin and China. "We have always used our standard DCS platform for electric utility projects over the last 25 years," adds Mark Converti, Honeywell Process Solutions (www.honeywell.com) Power Generation marketing manager.

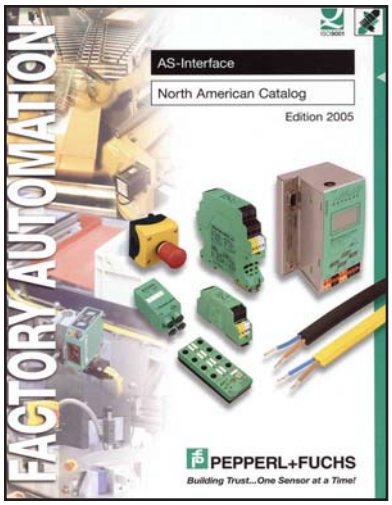
Control systems for coal plants are multi-million-dollar systems. Honeywell, for example, is installing a $10 million control system in China. And they are being built by big, solid companies. "Most new coal-fired plants will be built by investor-owned utilities that can shoulder the large financial risk (up to $1.5 billion) and long lead time to design and build large coal-fired generating units," says Leimbach. That would be a good company to work for, one not likely to lay you off.

Editor's note: *We have published complete comments from vendors on controlglobal.com.*

Rich Merritt, Senior Technical Editor
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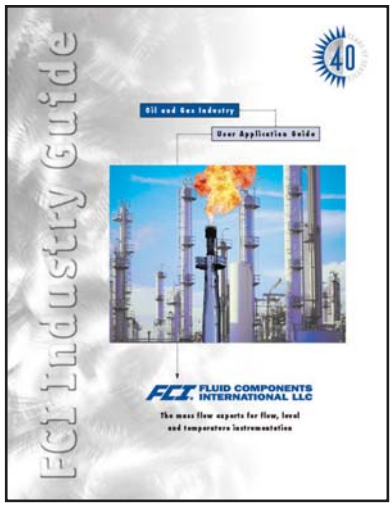
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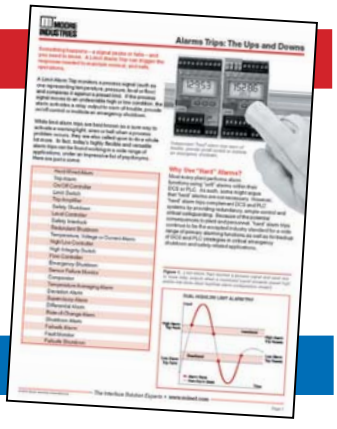
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People for Process Automation

A Recipe for Batch Automation Success: S88

It's been 10 years since the promulgation of S88 and, as George the elder used to say, it seems appropriate at this "juncture" to take a look at how it's faring in the process industries. But first, a little background.

Batch has been described as one of the oldest forms of process manufacturing, and over the millennia it evolved to include all types of control technology. Although automation controlled temperature and pressure, valve actuation, and motors, the the operator remained paramount and central to batch processing—unlike continuous processes, where the role of operator turns to monitor once things get going.



"The S88 series of batch standards is the most successful ISA standard ever produced. Why? Because batch and discrete control covers a huge area of industrial applications."

By the early '80s several leading process control companies began to offer truly automated batch systems commercially. With each offering a proprietary view of batch control, there was no generally accepted approach to automating processes. Training operators was a problem, systems integration became a customized nightmare, and everyone was speaking a different language to describe essentially the same thing. With everything so specialized and proprietary, automated batch control systems were as hard to sell as they were to buy and implement profitably.

In 1988, a committee sponsored by the Instrument Society of America (ISA) was formed to find a way through this FUD-filled swamp. Called SP88 (for Standards and Practices Committee number 88) it took the committee until 1995 to publish ISA-S88.01-1995 "Batch Control– Part 1: Models and Terminology." Made up almost equally of control system designers and end users making chemical, food and pharmaceutical products, the SP88 committee met every two months for more than five years working through the issues and defining the problems associated with batch production. What emerged was a compilation of principles and practices that apply equally well to manual control and to automated processes.

S88 champion and Manufacturing Automation Associates senior consultant Lynn Craig has spent the better part of his 30-plus year career herding the batch control cat across the industrial prairie. He's overseen the rise of S88 and finds that as of 2005 it is the standard which many in the process industries embrace. "In the control automation community, S88 is broadly accepted," says Craig, "especially by the pharmaceutical and specialty chemical industries."

Dr. Maurice Wilkins, World Batch Forum chairman and head of process automation at Millennium Specialty Chemicals, a Lyondell company, bears witness as well. "The S88 series of batch standards is the most successful ISA standard ever produced. Why? Because batch and discrete control covers a huge area of industrial applications—from blending operations in a refinery to pharmaceutical manufacturing and from chemicals to consumer products."

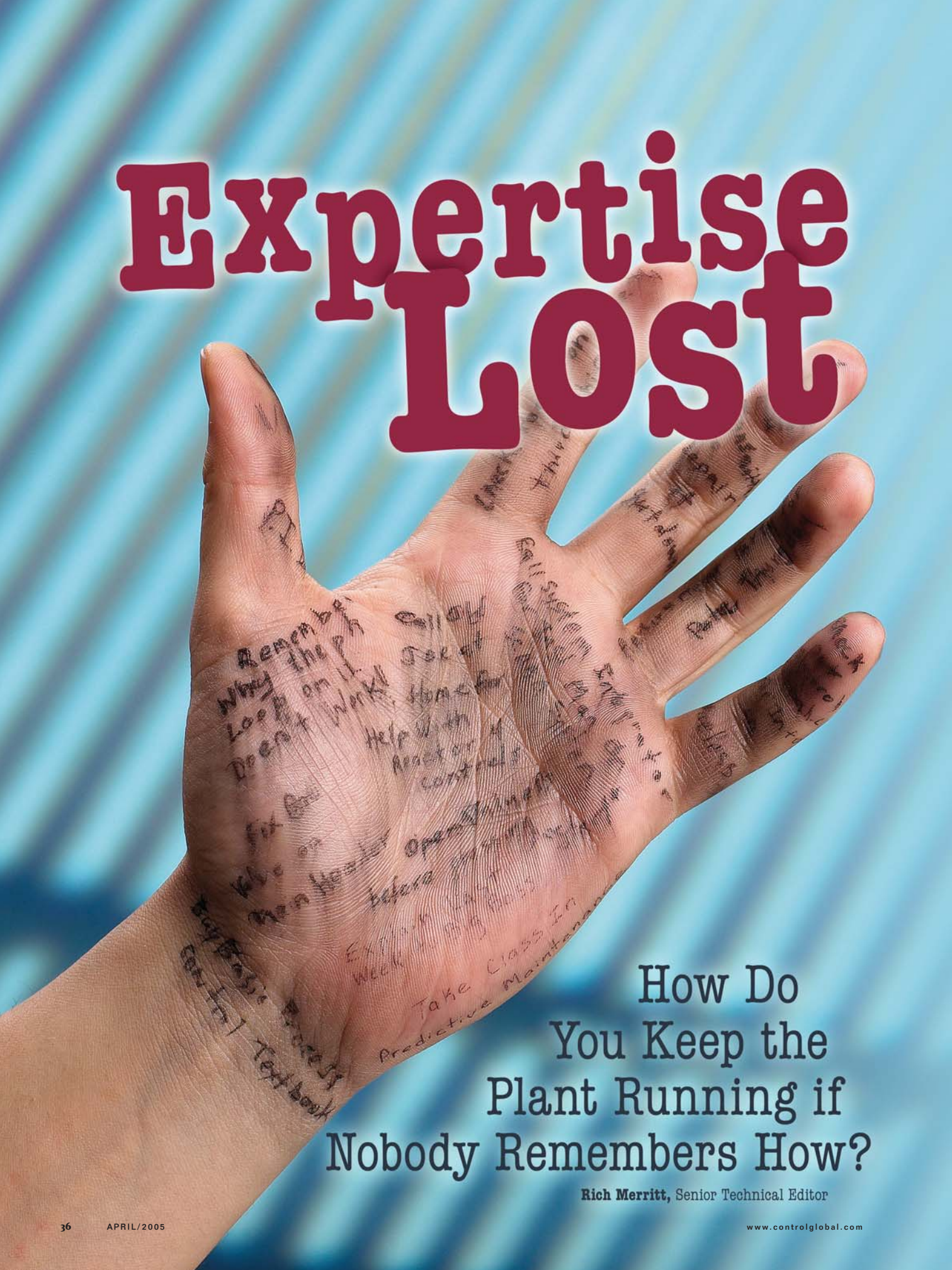
Wilkins says S88 has retired the "geek" –you know, the lone eccentric at the plant, schooled in the black arts, and the only one who knows how to get the automation systems to work right. "In 1988," says Wilkins, "I visited a chemical plant in England where they were looking to replace a batch system. As it existed, the system had three recipes, each with 14,500 lines of code and the indispensable geek." But, thanks to S88, he explains, that has all changed. "Aimed at making batch process control easy for operations and production people," says Wilkins, "it modularizes the plant, as well as the procedures and control functions so that a change in recipes or a revision in the process doesn't require a geek, a system rebuild or a reboot." Gee, no wonder it's growing in popularity.

The definition of "function" is the key says Craig. "The nice thing about dealing with functional definition is that it doesn't matter how the function is carried out. Material addition, for example, is exactly the same function whether carried out by automated equipment or by a person emptying bags into a tank. The details of how material is added are engineering issues."

According to Craig, ultimately the standard will encompass five parts. But, so far, it is parts one through three that have been published: Batch, Data Models and General Recipes. Parts four and five are currently in committee and include Managing Data from Process Models and Standards for Integrating Third-party Technologies, respectively. How's it going? "Part four looks like it is very close, and five is coming along—but it's a lot like watching snails play chess," jokes Craig. "These things can take forever."

Steve Kuehn, Managing Editor
skuehn@putman.net

Expertise Lost



How Do You Keep the Plant Running if Nobody Remembers How?

Rich Merritt, Senior Technical Editor

Layoffs, cutbacks, outsourcing, retirements, forced retirements, mergers and consolidations are stripping plants of instrument engineers, control engineers, technicians, maintenance people and operators. Once gone, their accumulated knowledge of the plant's processes goes with them, leaving companies without in-house expertise. Companies are turning to vendors for help.

If current trends continue, most of you will no longer be working as a control engineer for an end user company 10 years from now. More likely you will be working for a systems integrator, a process control vendor, or a service company of some kind. Soon, there may be nobody left in the plants, your accountant-bosses will have subcontracted all the work out, and process control will have become another purchased utility, like electricity and water.

That's what happened to the engineering and technical staff who worked at the Danube Refinery, Százhalombatta, Hungary. The refinery is the biggest production facility of the MOL Group, a huge oil conglomerate in Europe. Eight years ago, 130 engineers and technicians worked in the refinery. Today, after downsizing, outsourcing and restructuring, all of them are gone.

"As the former instrumentation department disappeared, most of its services were taken over by vendors like Honeywell," says an MOL manager, who asked to remain anonymous. Another 45 employees were eliminated when the refinery contracted out operation of its wastewater and waste incineration facilities to EarthTech in the UK.

The refinery also gave Emerson $10 million to design and install a new control system as the main automation contractor (MAC) for a control project. As the MAC, Emerson will perform front-end engineering design, applications engineering and project management, install and commission the automation solution, and supply operator training.

That doesn't leave much work for the refinery's control people. "The company tries to hire back or subcontract former MOL employees when it needs specific knowledge," says the anonymous manager.

This is happening at smaller companies, too. "Our chief instrument engineer left for greener pastures, and was never replaced," laments a manager of a small, specialty organic chemical manufacturer in the Midwest. "Then another instrument specialist left. Now we are relying on an engineering service to support our PLC programming, HMI programming and instrument specifications, and relying on process engineers to tune controllers. The few remaining instrument technicians we have do not have the time to tune controllers since there are too many other fires to put out."

The power industry is an excellent example of what happens when knowledge leaves. As we point out in this month's Control Report (p31), no coal-fired power plants have been built in 25 years, all the people who knew how to control coal-related process have retired, and there is no one left–except a few vendors–who know how to control a coal-fired power plant. Many in the process industries are going down that road, too, as knowledge leaves the plants.

One reason for loss of institutional knowledge is the aging and shrinking workforce in our business (Figure 1). A CONTROL reader poll shows that 71% of you are 41–50 years old, and 12% of you are over 50. Only 3% of you are under 25 and ready to step into their shoes and learn the business. Few young engineers choose to enter our profession, maybe because their fathers warn them away. Or maybe young engineers avoid control because of the disturbing trend of process companies to contract out major parts of a control engineer's work and shut down entire instrumentation departments, as MOL did.

A Boon to Suppliers

There's no question that the loss of knowledge in plants is benefiting suppliers, such as process control vendors, systems integrators and service companies. "We are seeing a very large upswing in engineering services," says Bill Robertson, director of global services and support for process systems and solutions division of Emerson Process Management (www.emersonprocess.com). "Customers are increasingly outsourcing engineering services on new projects, migrations and, to some degree, ongoing system maintenance and operations."

Honeywell says it takes deep pockets for a process company to keep up these days. "Unless a plant has the luxury of being able to overstaff and over-train, they will have to address the issue through the strategic use of suppliers who combine expert knowledge with technology that allows that knowledge to be disseminated out over a broad customer set," says Andy Drexler, Global Marketing Leader, Services,

Honeywell Process Solutions (www.honeywell.com). Drexler, of course, means companies like Honeywell.

"To prevent the impact of lost expertise, plants must carefully select partners whose expertise they can use on an 'as needed' basis and who offer the technology to help less experienced workers quickly reach a level of proficiency that once required years of experience," he adds.

All the big process control companies are aware of the outsourcing trend, and are gearing up to provide services ranging from design to installation to operations to maintenance. John Berra, president of Emerson Process Management, agrees that Emerson is capable of doing everything in a plant, including running it. "We have no plans to offer such a service," says Berra, "and I doubt that any company would want us to actually run their plant."

Nevertheless, the vendors are developing new hardware and software that gives them the capability to take over more and more responsibilities, such as automated monitoring, documentation, tuning, diagnostics and so on.

"Advanced diagnostic capabilities, such as our Loop Scout /Alarm Scout Service, enable the Experion control system to quickly identify and isolate problem areas in the large sea of control loop and alarm performance data," says Drexler. "These services automate and optimize plant work process that would have required significant time and effort from expert personnel in the past."

Emerson just announced its CSI 9210 machinery health transmitter (Figure 2), which automates rotating machine diagnostics. The system predicts when a motor-pump set needs maintenance. "Analysts who know how to interpret vibration data are leaving the plants," says Brian Humes, VP of Emerson. He says they are retiring or forming their own engineering companies, and the plants are not hiring anyone to take over the jobs. Their knowledge is not being passed on. "The CSI 9210 performs the necessary analysis and identifies the problem, so maintenance can take care of it without an analyst."

Unintended Consequences

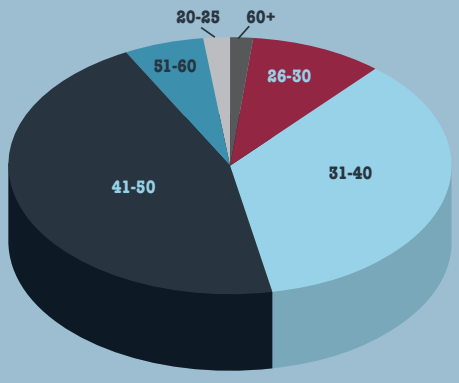
No one can blame vendors for taking advantage of the opportunity. Plants are losing or divesting their control systems professionals for various reasons, some good, some bad. Sometimes the motivation is a clear attempt to shake up and modernize plant and organizational systems that are stuck in the mud. Sometimes the motivation is clearly to appease Wall Street analysts. Sometimes, the motivation is just plainly to provide more income for the top management of the company. Companies do what their management believes they have to do. And so we meet the law of unintended consequences.

In many cases, plants that divest themselves of engineers have nowhere else to turn for help but to their vendors. The vendors have the necessary expertise, experience, process knowledge and a staff of engineers. However, rumors persist that the vendors are stretched very thin these days, and are not keeping up.

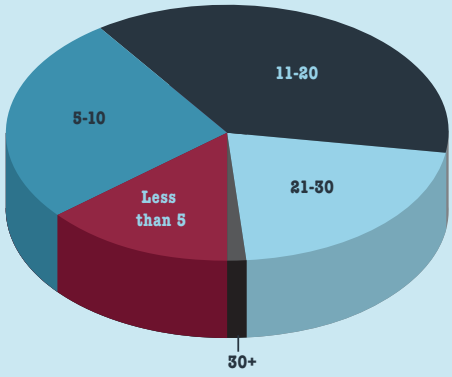
Glenn Givens, a control systems specialist at Innovention Industries (www.innovin.com) in Burlington,

Figure 1: Aging, Shrinking Workforce

What Is the Average Age of People Who Work in Automation at Your Plant?



What's the Average Length of Time (In Years) People at Your Plant Have Worked in Automation?



The aging, shrinking population of control engineers is not being replaced by younger engineers.

Source: CONTROL Magazine reader poll, www.controlglobal.com.



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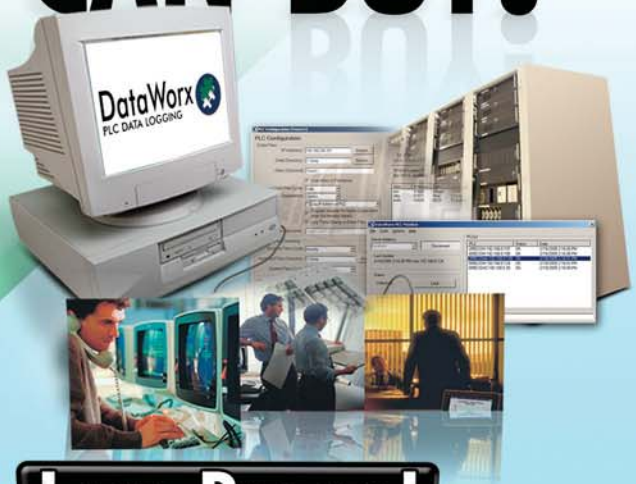
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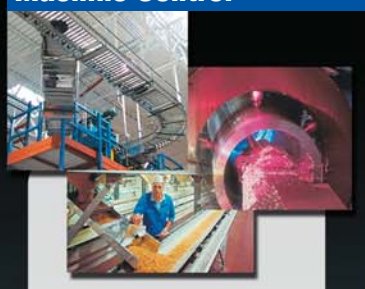
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Built-in comm ports	2	2	1
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Ontario, Canada, certainly isn't impressed by vendor expertise. "If you ask one of their technical people a question, they usually know less than I do and their answer is 'I don't know, why don't you try it?'," says Givens. "When I find an important bug and I have hard data, they first say they're interested, and then nothing is done. I don't bother telling them any more. I see their controls expertise as 'entry level' or perhaps slightly more advanced than the average instrument technician."

Figure 2.

Vibration Analyzer in a Can



Analysts who understand vibration data are leaving plants for greener pastures, and are not being replaced. Emerson's CSI 9210 machinery health analyzer diagnoses problems with rotating machinery and tells operators and maintenance exactly what is wrong and how long before the motor-pump fails completely.

The manager of the small specialty chemical company says help eventually comes. "When we scream for help, so far we have been able to get help from our engineering service," he says. "But it comes at a high price, and sometimes with a delay."

To keep from getting overwhelmed by the increasing requirements of plants, the vendors are coming up with ways to simplify their lives.

"We've systematically engineered complexity out of our systems, at times eliminating tasks that used to take up valuable engineering time, like configuring highway addresses," says Robertson. "Advanced control is now easy enough for the basic automation engineer and some configuration tasks can be accomplished by operators instead of engineers."

Berra says they've seen this trend at Emerson. "As our diagnostic tools do more of the low-level work, we've been moving up our engineers into more responsible jobs," he says. "Our support engineers spend more time on bigger customer issues, and less time diagnosing equipment problems."

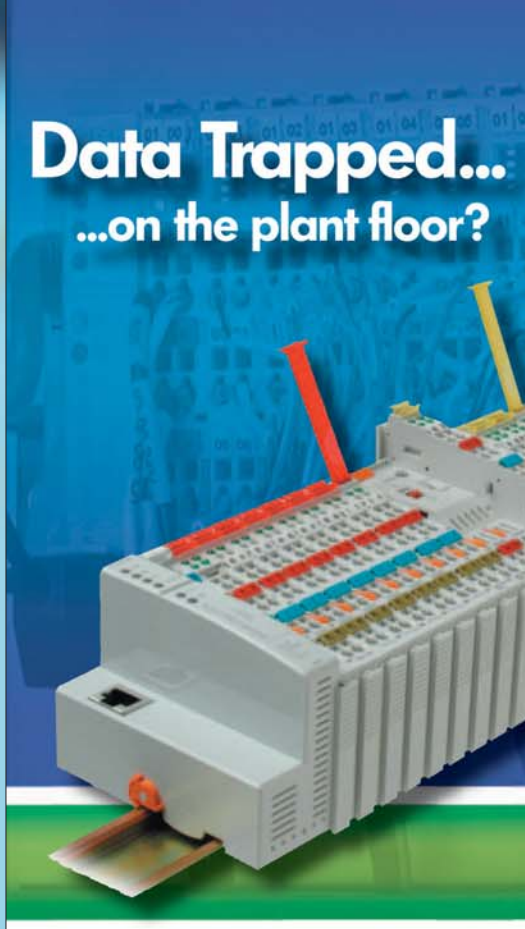
Perhaps such a realignment of work is the future for control engineers. Instead of working with the actual equipment, tuning loops and diagnosing problems, you will let software do that work while you operate at a higher level–such as approving invoices from vendors.

Keep the Knowledge

When experienced engineers leave the plant, they take valuable knowledge with them, such as the reasons for certain types of loop-tuning parameters, why PLC logic was done a certain way, or why batch sequences are done in a particular order. Therefore, you should make every effort to capture that knowledge before it is sent away. Good documentation right from the git-go helps, of course, but if that's lacking, you need to create it. To justify the expense of documenting your existing system, you can explain to your senior management that such documentation will reduce the cost of future upgrades.

"Often the lack of knowledge of existing systems becomes obvious when customers attempt to migrate to newer generation equipment," says Robertson. "The first and sometimes painful step along this path is to fully document what the customer has. We provide a wide array of services and technology to make the process of untangling the spaghetti code that we occasionally find for the customer so they can migrate their valuable configurations."

Givens has known this for years. "We configure and build controls, write software, diagnose process problems, tune



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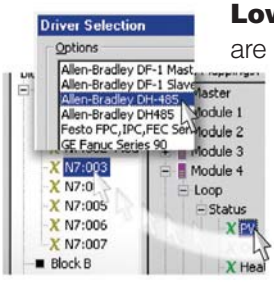
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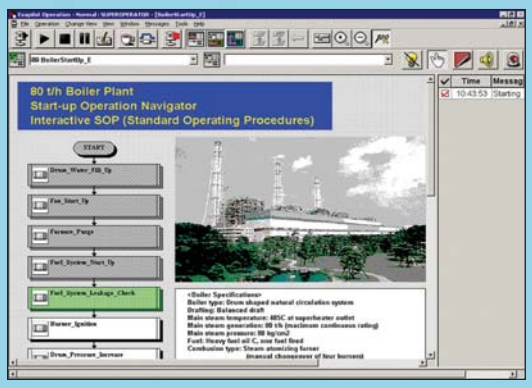
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controls and, occasionally, train technicians," he says. "We have detailed records of every single bump test done to every single loop since the day we went into business. I can find the data files for every one and tell you how I calculated the tuning parameters, how I determined how much backlash/stiction was present, and so on."

Figure 3.

Capture That Knowledge



SOURCE: YOKOGAWA

Yokogawa's Exapilot allows operating personnel to standardize and automate what would normally be manual procedures, incorporate the know-how and skill of expert operators, and document everything.

A great deal of new software is available to help you document your system. Wonderware's Industrial Application Server, for example, allows control and system engineers to standardize system implementation and encapsulate knowledge on ArchestrA-based systems, says Steve Lewarne, Wonderware's vice president of product marketing. "ArchestrA provides a structured model and approach to designing supervisory automation and information systems," he explains. "ArchestrA is also self-documenting with respect to initial configuration and modifications made throughout the life cycle of the system. These allow system engineers to easily design, build, deploy, and maintain secure and sustainable automation and information applications."

Yokogawa has something similar. "Yokogawa has developed a software solution called Exapilot (Figure 3) for capturing this fleeting best-operations knowledge," says Fred Woolfrey, Productivity Solutions Consultant at Yokogawa (www.us.yokogawa.com). "Exapilot is a Microsoft Windows-based operation efficiency improvement software package that allows operating personnel to standardize and automate what would normally be manual procedures and incorporate the know-how and skill of expert operators. Operating personnel can build and modify these procedures with an icon based interface. Users can easily configure and test complex procedures. Procedures built with Exapilot can be used to standardize and automate manual procedures, improve plant efficiency, and improve the safety of plant operations."

Because vendors must document what they do when developing and configuring control systems so they can support them later, they've developed such tools for their own use. Check with your control system vendor to see what's available.

Train and Keep Maintenance

An I/E Supervisor at a power company in Texas trains his technicians and spreads the knowledge around, so he's not crippled if a key person leaves. He's also less dependent upon vendors. "When we purchase an upgrade or a new system, we add in cost for training in the price of the equipment," he explains. "If it is not cost effective to send all the techs, then the ones that do attend the training are required to train the rest of the group. I also rotate areas of responsibilities with the techs to keep them current in the understanding, troubleshooting and repair of the equipment. This helps these individuals become more of an asset to the company and it makes my life a whole lot easier. You can never train enough."

He loves the new open control systems, because it gets him away from proprietary hardware. "Control system manufacturers used to build and supply every

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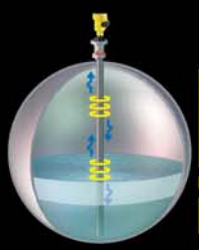
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component, but now they buy from third parties to build their systems," he says. "Believe it or not, this is really a cost savings to the customer. For example, if I lose a hard drive on my engineers' console, I run down to Radio Shack and buy one at a quarter of the cost than when you had to buy from a single source."

It's not just economics. He says the vendors have lost some of their expertise in repairing their own equipment. All the vendors responding to my question of how much they support their older equipment said they go back to the dawn of time, but that's what you would expect them to say. They did not say they had experts doing the work.

Therefore, you can probably build a case against letting a vendor take over maintenance on your system on the basis of cost and performance. Cruising the automation lists on the Web would probably get you enough anecdotal evidence to even make a flint-nosed bean counter think twice about outsourcing maintenance. Keeping equipment knowledge in-house helps keep jobs for you and your techs, but you also have to be cost effective.

Analyze Those Loops Yourself

The process control vendors and service companies have built powerful analysis and diagnostic capabilities into their new software systems. They have the ability to monitor every process variable, put all the data into a process historian, and analyze it on a loop-by-loop basis for efficiency from remote service centers or locally, in your plant.

You might want to take advantage of the same tools to preserve your job. PlantTriage from ExperTune and similar software from other vendors can perform the same service that the process control companies offer. In fact, the vendors might be using the same software, because it works.

"A performance monitoring package like PlantTriage prioritizes the loops that need attention and automatically diagnoses the problems," says John Gerry, president of ExperTune (www.expertune.com). "The only way to survive and be competitive is to use software tools like this." It also helps you preserve process knowledge, because it records all PID tuning changes and allows you to enter notes as to why you tuned it that way, he adds.

Run Your Own Server Farm

With server-based systems soon to be announced, the big control vendors will be able to run your plant from centralized server "farms," where specialists of all kinds will be available to diagnose problems, retune loops, decide when maintenance needs to be done, and even operate the plant from afar. In other words, the entire control system and all its engineering and tech support can be outsourced.

There is no reason why a forward-thinking control engineer such as yourself could not set up a similar server farm in your company; that is, a centralized system that monitors and controls all your company's process operations around the world. In other words, instead of having instrumentation departments in each plant, set up a central Instrument Engineering department at company headquarters, and run all your plants from there.

Such a system documents and preserves all the process knowledge in one place. A centralized system also preserves the jobs of your company's expert control engineers, techs, and operations people, and keeps all their knowledge in house.

The wave of the future is to consolidate all the control, maintenance and diagnostics work in a plant anyway, so just ride the wave, set up your own central operation, and keep your knowledge in house. Otherwise, you might do better to shop your resume to whoever will have to pick up the ball as the knowledge disappears from your plant.

Adversity always breeds opportunity. When your business accounting management starts looking for ways to replace engineers and techs with outside services, show them that you can do the same job at a lower cost by using the same knowledge management tools that the outside suppliers will use if you don't.



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PATENT INFRINGEMENT CASE REMAINS A TANGLED WEB

Solaia May Have Finally Met Its Match Now
That It is Facing Rockwell Automation in Court **By Steve Kuehn, Managing Editor**

If you exclude sex and murder, this increasingly complex drama has a plotline that just about covers everything else involved with human avarice including suspect ethics, pride, jealousy, intrigue, conspiracy and betrayal. No it's not *The Sting*, but to many in the process and manufacturing industries it's beginning to feel like one. Actually it's the continuing saga of a patent infringement dispute over the ubiquitous way most companies manage control automation data.

To recap, in early 2001, Solaia LLC purchased the rights to the now controversial patent known as the " '318 patent" from Schneider Automation. Soon thereafter, it engaged the services of renowned Chicago patent enforcement law firm Niro, Scavone, Haller & Niro, to pursue its claims against users of the technology defined by the patent granted in 1991 to Square D (acquired by Schneider) and employee and inventor Brooke T. Roseman. Formally, it's Patent 5,038,318 covering the "Device for Communicating Real Time Data Between a Programmable Logic Controller and a Program Operating in a Central Controller." In a twist that is becoming a hallmark of contemporary, high-stakes patent enforcement litigation, instead of going after companies (at least initially) that make the technology that purportedly infringes on the '318 patent, namely Rockwell Automation, Siemens Energy & Automation, and GE Fanuc Solaia targeted companies that use computer controlled factory automation systems—which just about covers everybody who makes something.

Regardless of the legal arguments most users of factory and process automation use the technologies covered by the '318 patent. The stakes are extremely high for vendors, but if Solaia prevails it's the users who may suffer in the long run.

Choice the First Victim

The freedom to choose automation weighted towards technical merit could be the first victim. Users may not venture to purchase a certain technology because it might be vulnerable to a future patent infringement claim, even though its implementation might benefit the organization. As editor in chief Walt Boyes points out in his editorial "The Death of Innovation?" (CONTROL, Feb. '04, p9) "It is important to remember these cases are about what … users will be permitted to use in hardware, software and networking technologies in the future." Price is another aspect, because if suppliers are forced to pay newly minted royalties on firmly established automation technologies they are likely to eventually pass-along the costs to users. Similarly, users paying royalties on technologies they've banked their entire production line on will also have to pass on or redirect such costs as well. Such intrinsic costs are truly a hidden tax that create a damping effect on the healthy economic dynamics of industry in this country and put further pressure on the U.S. to remain competitive with countries that do not maintain similar legal labyrinths.

PATENT LAW PREDATORS



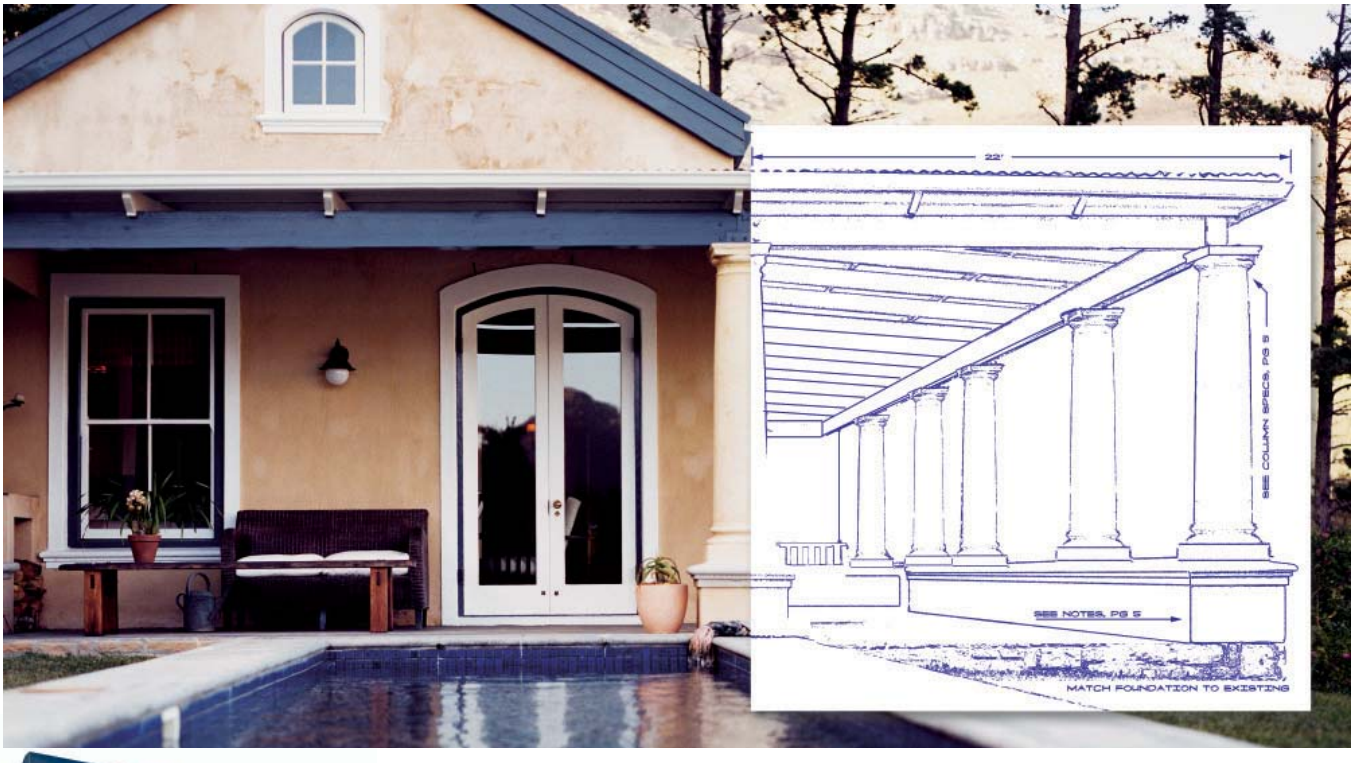
The nature of patent law has lead to predatory practices that can snare even innocent companies.

Boyes argues that the pace and volume of technological change and innovation has accelerated to the point were the patent office does not have the ability to correctly judge the true technical veracity of emerging technologies and has not served the automation community, nor U.S. commerce well at all by granting such broad patents (see "Technocrats Take the '318 Patent to Task").

In the case of the '318 patent, Schneider ostensibly chose to sell the patent and leave it to Solaia to invest the resources, shoulder the risk (as well as bad PR) and reap the rewards as they pressed their claims of infringement against it. But their involvement didn't end with a simple sale. Although at the time Schneider was loath to discuss the actual terms of the sale, it was eventually revealed that a percentage of the revenue generated by the '318 patent's licenses return to Schneider as part of the patent purchase agreement.

This connection has become quite a source of rancor between Rockwell and Schneider who remain direct and fierce competitors in the automation industry.

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Hot Summer

By the summer of 2002 the number of companies being drawn into litigation began to mount. In July of that year Solaia and its law firm filed against 16 more companies that ostensibly use Rockwell's equipment. These included ArvinMeritor (a former Rockwell International company), Boeing, Borg Warner, Callaway Golf, Chevron Texaco, Conoco., Eastman Kodak Co., Eli Lilly, Enbridge, General Dynamics, Gillette, Kellogg, Lexmark, Shell Oil, Sun Chemical and Tyco.

With such a clear threat presenting itself to its customers and its primary market, Rockwell's next legal move was as brash as it was unusual. In December, 2002 Rockwell filed a lawsuit alleging Solaia, Niro, Scavone, Haller & Niro and Schneider of conspiring to ruin Rockwell's reputation with its customers (Rockwell Automation, Inc. et al, v. Schneider Automation, Inc. et al,. (02-C-1195, E.D. Wis.). This pending case asserts claims of tortious interference and civil conspiracy, and violations of federal antitrust and unfair competition laws.

What makes it unusual is the fact that it names Solaia's legal counsel as a defendant in the suit. "This is an extremely curious strategy," says Thomas J. Donovan technology patent litigation expert and partner in the Chicago office of Barnes & Thornburg LLP. "It certainly is a bold move, and I'm sure they considered it very carefully, but it's risky. Suing the law firm makes it personal between the lawyers. It also, at least so far, seems to be counter productive. I would expect that this move has resulted in legal fees that are very substantial and not at all related to whether the Solaia patent is valid and infringed. Not to mention that judges don't tend to be fans of this type of litigation between law firms."

Responding, Niro, Scavone, Haller & Niro returned the favor and filed a countersuit against Rockwell and its law firms Howrey, Simon, Arnold & White and Fish and Richardson for similar transgressions.

While it may be a stretch to prove civil conspiracy *per se*, there is evidence that Solaia and Schneider are acting in concert to generate both a competitive advantage and a profitable return on what industry sources maintain is a very weak and overly broad patent. According to court order 03 L 794 from the Circuit Court of Cook County (dismissing a defamation and libel suit brought by Solaia and Niro, Scavone, Haller & Niro against *Start*, a manufacturing industry magazine that reported extensively on the case) "Solaia purchased the patent (known as the '318 Patent) from Schneider for $1.00 plus a percentage of any monies recovered through enforcement of the patent." *Start* Magazine's coverage of the story also offered information

GE FANUC SETTLES WITH SOLAIA

On March 9, GE Fanuc Automation Americas, Inc., announced it had signed an agreement with Solaia Technology LLC, "which provides broad rights to GE Fanuc Automation, GE, FANUC Ltd of Japan, and their customers under the Solaia 5,038,318 patent."

According to GE, Solaia releases any potential claims on the '318 patent against customers for the purchase or use of products, software, services, equipment or devices made or supplied by GE Fanuc Automation, GE or FANUC LTD. Furthermore, says the announcement, any system that was substantially integrated by a GE entity is released, regardless of which company manufactured the individual system components.

"GE Fanuc reached this important agreement with Solaia in order to assure our customers that they have no potential risk for litigation under the '318 patent," said Jeff Garwood, president and CEO, GE Fanuc Automation. "At this critical time in our global economy, manufacturers need to concentrate on improving productivity, reducing costs and achieving peak performance, and we are committed to helping our customers do just that with best-in-class automation solutions and the worldwide support of GE."

Return on Investment.

"We utilize 3 DCS vendors within the company," says Nagórko. "We began with each vendor running approximately 30% of our control. ABB provides an easy evolution path to its new products and to extend these systems. Even a step change to system operation does not require replacing hardware to take advantage of the latest software developments. Not all vendor products permit that kind of compatibility. ABB now controls over 60% of our processes."



Waldemar Nagórko, Manager, Process Automation Department ORLEN PKN

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from court memoranda that the agreement between Schneider and Solaia details a list of 136 key customers with grant-back non-exclusive licenses.

No Really, It's the Money

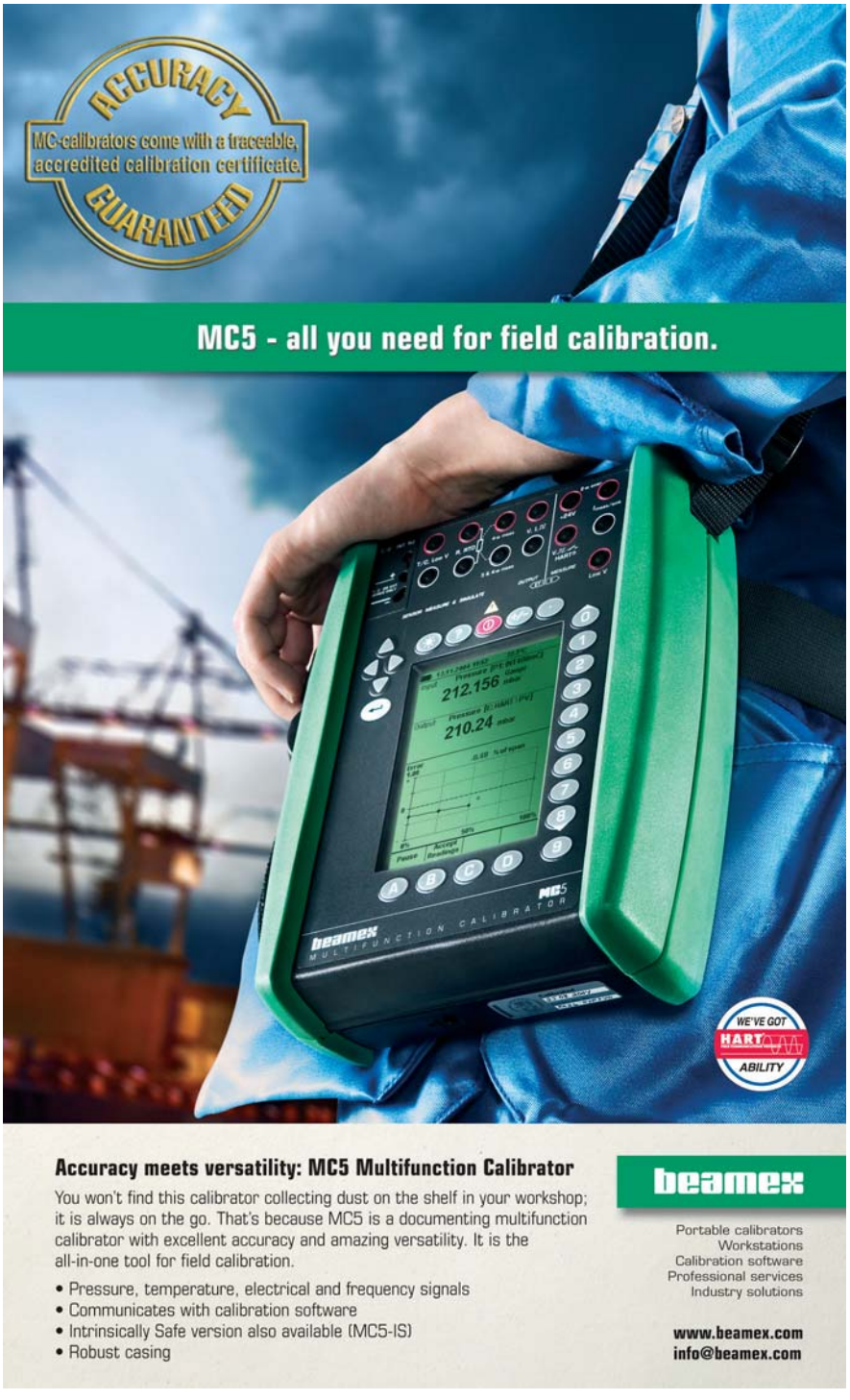
In his Nov. '04 *Wall Street Journal* story "Patent Dispute Embroils Host of Industries," Timothy Aeppel reports Ray Niro, Solaia's attorney as saying the firm has secured more than $24 million in license fees from some 60 U.S. companies. Initially, Solaia's demand letters asked for 1/2 of 1% of the recipient's output for two years, and originally capped at $600,000. Industry sources say that figure has now been halved to $300,000. "Considering it may cost anywhere from $1–2 million to sustain a protracted legal defense against an infringement claim," says Barnes & Thornburg's Donovan, "it's easy to recognize the strategy at play here."

Yep, it sure is. Press an infringement claim that's hard to disprove and set the settlement price at a point where prudent counsel recommends settling, because when the dust settles, you might not only be paying the license fee, but also be stuck with a couple of million in legal fees to pay for your attorney's efforts.

"For an accused infringer in cases like this, it is a lot like a high-stakes game of poker—even if you think you might have an invalidity defense," remarks Donovan, whose engineering and law degrees and almost 20 years of intellectual property litigation experience give his opinion some weight. "Until you get to court, you don't know all the cards the opposition might have to play against you. And, under the patent laws, patents are presumed valid. What that means is that if the validity of a patent is challenged, the patent owner does not need to prove its patent is valid, rather it is the accused infringer that has the burden to prove the patent is invalid. Compounding that is the fact that the more settlement licenses Solaia obtains, the more credence it lends to the validity of the patent in question."

What was troubling many observing the case two years ago was that, as it gained momentum, the tactic of demanding licensing fees via threat of lawsuit, but setting them well below what it would cost to mount a realistic defense was starting to pick off companies one-by-one.

Continued on p. 55



TECHNOCRATS TAKE THE '318 PATENT TO TASK

Considering the highly developed nature of patent law, most of the patents granted are likely to be legitimate. According to U.S. Dept. of Justice, Bureau of Justice Statistics, the number of patents granted has increased 197% from 62,000 annually in 1983 to 184,000 in 2001. But with so many patents being granted for things not envisioned when the laws were devised (Internet commerce for example) it's equally likely that there will be some that truly aren't. Many agree that this is true for Solaia's 318 Patent.

In a letter from a person identifying himself as an "industry veteran who prefers to remain nameless," (Start magazine "Mail Call," Jan. '03, p10)." The '318 patent "basically patents a method for controlling a manufacturing operation using a spreadsheet add-on that talks directly to the registers in a programmable logic controller (PLC) without using the operating system or any I/O device." He points out two big problems with '318: One, it's extremely narrow, and two, it had to be narrow to avoid a rejection based on a considerable amount of prior art—something that anything patentable has to rise above to be granted.

Mr. Nameless' appraisal is succinct, but for a comprehensive analysis, the staff at High Tech Systems (HTS), a North Carolina-based systems integrator provide a detailed appraisal on their website (www.htservices.com). High Tech Systems, founded by president Keith Baldwin, (a founding member of the CSIA), recommends that to better understand the problematic nature of the '318 patent it's instructional to understand what is patentable.

New and Useful

The U.S. patent office www.uspto.gov) offers this: "In the language of the statute, any person who 'invents or discovers any ***new and useful*** process, machine, manufacture, or composition of matter, or any ***new and useful*** improvement thereof, may obtain a patent,' subject to the conditions and requirements of the law. The word 'process' is defined by law as a process, act or method, and primarily includes industrial or technical processes. The term 'machine' used in the statute needs no explanation. The term 'manufacture' refers to articles that are made, and includes all manufactured articles. The term "composition of matter" relates to chemical compositions and may include mixtures of ingredients as well as new chemical compounds. These classes of subject matter taken together include practically everything that is made by man and the processes for making the products."

The HTS analysis then quotes Roseman from the original patent application on a number of points but one stands out: "The invention furnishes a program or instruction means that operates through a general purpose spreadsheet program to effect information transfers to and from the addressable registers of a PLC and assigned cells of a displayed spreadsheet directly from the spreadsheet program and without transfers through the operating system program or a specially written device driver program."

Says HTS, "We don't see how using the specific words 'PLC' or 'spreadsheet' constitutes anything 'new' or 'improved.' " The systems integrator's analysis argues that "PLC" and "spreadsheet" are nothing more than a type of computer and type of software that are communicating with each other.

"There were other companies already making SCADA software that talked with PLCs at the time," says HTS, and "that because computer programs were already talking to PLCs, uploading and downloading data, one might think that a claim about PLCs and spreadsheets would not hold up."

The discussion of prior art doesn't end there either. According to HTS, people were already reading and writing data to/from real-time controllers and PLCs using databases. "In fact, our records show that we were using an IBM PC to read and write databases in factory automation in 1984. [And HTS] worked with a group at Texas Instruments in Johnson City, Tenn. in 1985 that developed a system tying manufacturing databases into automated lines, controlled by PLCs."

Wrapping it up with a bow, HTS declares: "Mr. Roseman's invention was to interface Lotus 1-2-3 directly to a Square D PLC. We can understand—but do not agree with—a claim against anyone using Lotus 1-2-3 to read and write [to] Square D PLCs. All other claims seem completely unjustified."

Eliminating Risks.

"There was concern that we would have to idle portions of the plant while we tied in to the loops" said Vogler. "With the help of ABB's technicians, pairing up with our technicians in-house, they found a way to combine the loops without ever having a unit outage. We never took a unit out for the sake of making this upgrade project happen."



Mark Vogler,
Manager of Engineering
Citizens Thermal Energy

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Intellectual Property Rights

Considering the highly developed nature of patent law, most of the patents granted are likely to be truly legitimate. But with so many patents being granted for things not envisioned when the laws were devised it's equally likely that there will be some that shouldn't have been granted in the first place. Many agree that this is true for Solaia's '318 patent.

"Regardless, in almost all instances, a defendant accused of patent infringement contends that the patent is invalid," says Donovan. "Companies certainly have the right to protect their patents and secure compensation from infringers and those who wish to use patented technology. On the other hand, a defendant who believes a patent should not have been granted certainly has the right to try to invalidate the patent. But, until that happens, and until a court holds the patent invalid, the patent will be presumed valid."

As the new suits and counter suits began to fly in 2002, many individual companies were settling out of court. Those getting to court such as Clorox, were settling pretty quickly once they got a good look at what they were up against.

Up until that time, it appeared companies weren't aligning themselves with others to support a real challenge to the patent. However, throughout 2003, companies began to band together and it finally started to look like there would be enough legal resources to bring this thing to a head and decide, once and for all, if '318's broad claims are truly valid.

But Solaia's campaign to support its legal rights continued and in June 2004, *Solaia Tech LLC v. GE Fanuc Automation*, was filed in U.S. District Court for the Northern District of Illinois. This suit was one of the first infringement claims against a deep-pockets automation vendor. But instead of fighting it, GE Fanuc and Solaia settled early in 2005 (see GE Fanuc Settles With Solaia). However, in spite of what some might consider as GE Fanuc's capitulation, more than 16 individual companies united their legal effort and now have patent infringement cases pending with Solaia LLC in Illinois and Minnesota courts.

Furthermore, according to Rockwell Automation's 10-K in its 2004 Annual Report "In a suit filed by Solaia on July 2, 2002 in Chicago, *Solaia Technology LLC v. ArvinMeritor, Inc., et al*, (02-C-4704, N.D. Ill.), Solaia accused sixteen companies of infringing the '318 patent. We made arrangements with ArvinMeritor, which now owns and operates our former automotive business, to undertake ArvinMeritor's defense of Solaia's patent claims to seek to assure that Solaia's infringement claim against ArvinMeritor could be finally and actually adjudicated in the Chicago patent suit. In that case, Solaia responded on May 12, 2003 by suing us directly for patent infringement, demanding material monetary damages."

In the Minnesota case, which seeks a declaratory judgment against Solaia, the judge recently ruled against Solaia's motion to dismiss. Essentially that means the judge found that the 16-company group have a strong enough case against Solaia to continue its litigation and prove baseless the infringement claims and invalidate the '318 patent.

According to Rockwell's in-house intellectual property counsel John Miller, discovery is complete and dispositive summary judgment motions are pending on the key, critical elements of the suit, among them, primarily, that ArvinMeritor, Rockwell and its customers do not infringe on the '318 patent. According to Miller all appropriate and germane documentation supporting the motions has been filed and summary judgement is imminent. In fact, as you read this, judgments may already have been handed down, finally bringing the curtain down on this sordid tale of industrial intrigue and legal slight-of-hand.

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POWERFUL PLCs INCREASE PRODUCTION

Satisfaction Guaranteed, From IT to Maintenance

By William Love, Kredit Automation and Controls, Phoenix, Arizona

These days, powerful PLC-based control systems can handle most continuous, batch, motion or discrete applications. Fast PLCs are available with large amounts of RAM (2–6 MB). They can process large amounts of information in real time. These modern PLC capabilities can be leveraged to achieve higher production and better order processing and inventory management in a batch blending system.

LEGACY PRESERVED



The technology that replaced the original synthetic stucco batch process control system leveraged the existing PLC 5 investment by retaining the PLC 5 chassis and I/O. Key to the preservation: swapping the 5/25 processor card with a remote I/O card.

The batch blending system combines solid materials from silos with liquid ingredients pumped from totes in a mixer system. The process has some complexity, and various weighing and transport systems are involved, but mixing solids and liquids is the essence of it. The products this batch process produces are building materials (e.g., synthetic stuccos) that are sold in large pails.

A customer's order for a certain number of pails of a product causes an algorithm to use the formula for that specific product to create a set of steps referred to as a "batch." Any given step of the batch is either an addition of material, an automatic control action (e.g., turning on an auger), or a prompt for the operator to take an action and acknowledge the step.

Out With the Old

The batch system being replaced was controlled with an Allen-Bradley PLC 5/25. Because it had only about 25 kB of memory, the batch steps would not all fit in the PLC. Instead, the batch was created on a PC and constant handshaking had to take place to notify the PC to send the next step of the batch to the PLC.

Only one batch could be processed at a time. This was a fundamental design characteristic of the programming, not just a memory limitation. This meant a batch had to finish completely before another one could start—a very inefficient way to utilize the equipment. Worse, there are three mixers, and so only one mixer could be used at a time.

The primary goal of the project was to create a software system with six batch routines—two per mixer—that would enable all three mixers to be active simultaneously and let a second batch be active for a given mixer before the first was completely finished. This design would increase throughput by 25%.

Another important project goal was to provide a means to automate and integrate order processing, data logging, inventory management and any other IT function that might be added later. Some of these functions were handled by the new system; others were made possible by the "hooks" available in the new system to tie it to other computer systems.



And in With the New

The control system (see Figure) consists of an Allen-Bradley ControlLogix PLC, a computer (PC), and three PanelView terminals. The original PLC 5 rack and I/O were retained, but the PLC 5/25 processor card was removed and replaced with a remote I/O card. This was important because a large portion of the original hardware and installation investment was preserved. The new PLC has about 2 MB of RAM, is extremely fast, has Ethernet connectivity, and can be programmed in structured text as well as ladder logic. Ladder is a time-tested workhorse that provides a great troubleshooting interface.

Programming with Visual Basic

A new PC running Windows XP was loaded with Rockwell Software's RSView application development software. The traditional purpose of this software is to allow an engineer to develop operator interface screens but there is much more that can be done with it.

The system created with RSView is usually called a human machine interface (HMI) since operator screens are an important part of its functionality. A significant thing about RSView is that it comes with Visual Basic for Applications (VBA). A knowledgeable programmer can create algorithms for virtually any purpose and create rock-solid integration to other systems. For example, Kredit used VBA inside the HMI program to access the file system of the PC and read, modify and write files and folders on the PC hard drive. The object model of the HMI is available to the VBA code, including the tags, screens, alarm objects, trend objects, etc. Many of the HMI tags connect directly to the PLC tags, which means tight, fast access to the PLC is also available to the VBA code.

This is emphasized because it is quite different than creating custom programs in regular Visual Basic or C++ or a database package such as SQL Server. Those types of applications have issues that simply don't come up when using VBA inside the HMI. Issues pertaining to drivers, DLL files, operating system compatibility, etc., don't have to be dealt with. A new computer can be configured in minutes and, more importantly, a new computer can be configured with a minimum of or skills. Before this proj-

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2
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8
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ect, when the PC contained a custom program that worked with a database system, the plant's operators were not sure the PC could be duplicated. In other words, if the PC died, they were not sure they could restore the control system!

How We Did It

We were asked to create a control system that could accept a batch from any computer system running any operating system. The plant's current enterprise computer is an IBM AS-400 mainframe, but the design requirement was that the mainframe could be replaced with any other computer without requiring much if any reconfiguration of the batch control system. While that sounds daunting, the sheer generality of it suggested a simple method—simple if you leveraged the VBA in RSView.

The batch system we created was designed to use simple ASCII text files as input. Batches can be created locally by filling out a form in the RSView run-time application, or they can be made using Notepad or any text editor. The resultant one-line batch spec is placed in a text file in a specific folder. Independent of this process, an event-driven VBA routine checks the folder every few seconds and if it finds a file, it processes it. The beauty of this modular, decoupled approach is that it doesn't matter how a batch file gets placed in that folder—if it arrives there, a batch will be initiated. As a result, any computer with an operating system that can write a text file can initiate a batch if it has access to the right folder over the network.

For a while, batches were created locally. But the day came when the corporate computer would create them remotely. Corporate IT personnel were expecting a typically challenging project to tie their IBM system to a Microsoft system. They were pleasantly surprised to discover that they simply had to drop a text file into a folder. No changes whatsoever were required to be made to the batch system allow it to accept and process batches from the corporate IBM mainframe. The IBM programmers had no trouble programming their system becausee creating a text file is so easy.

The original control system required a proprietary program and a complex database application that was hard to set up, change or maintain. In fact, it was not known if all the software needed could be obtained and configured on a new PC. With the new system, installing RSView and copying the application folder is all that is needed. Truly valuable was the fact that an entire third tier of very complex and poorly understood software was eliminated.

Throughput Increased 25%

The new control system leveraged the existing PLC 5 investment by retaining the PLC 5 chassis and I/O, as well as all associated wiring. The new ControlLogix processor easily fit into the existing panel. From a wiring and installation standpoint, the retrofit was painless. The new PC was loaded with RSView and connected to the PLC via Ethernet. The VBA code rsesponsible for so much processing lives inside the Rockwell Software RSView application. No additional software, drivers or anything else is needed. VBA in RSView is a standard feature and does not require any additional software to function correctly. The application folder was simply moved from the developing engineer's computer to the customer's production computer and it was ready to go.

The new batch control system increased throughput by 25%. The highly modular software system that was installed is totally amenable to change and has allowed modifications and improvements to be made on an ongoing basis to continually optimize and improve operations. Order processing and inventory management functions were also automated and integrated with the control system.At the end of the day, everyone, from corporate IT to the local maintenance department considered the project a great success.

William Love has been in the systems integration business in Texas and Arizona for 10 years. Contact him at 602-315-2561; or bill@kreditautomation.com.



SCADA SYSTEM REPLACES AGING, PROPRIETARY DCS

More Than 450,000 Real-Time Variables From PLCs, Physical I/O Points and Serial I/O Points Provide Detailed "Information Anywhere" **By Duncan Dodds, Project Manager, Western Mining, Olympic Dam, Australia**

WMC Limited began as a gold exploration, mining and management company in Western Australia in 1933. Since the 1950s the company has diversified and expanded to become a significant producer and exporter of processed minerals and metals.

WMC began exploring southern Australia for copper in 1961. In 1977 drilling revealed a large prospective ore deposit located in the Stuart Shelf geological province at Olympic Dam (Figure 1). Mineral reserves are now known to be 11.4 Mt of copper, 0.34 Mt of uranium (as uranium oxide), 400 tons of gold and 2,790 tons of silver.

Between 1988 and 1995 production rates increased significantly and continue to increase. With increased production, came a need for further expansion which will be executed in two phases. The first increases the production rate from 85,000 tpa copper to 150,000 tpa with capacity for 200,000 tpa, and the second increasing overall production to 350,000 tpa.

A copper production rate of 200,000 tpa will yield average production of approximately 4,630 tpa of uranium oxide, 2,050 kgpa gold and 23,000 kgpa silver. Future expansion may generate production rates of 350,000 tpa copper, 7,730 tpa uranium oxide, 3,630 kgpa gold and 49,600 kgpa silver.

FIGURE 1.

OLYMPIC DAM MINE



WMC's Olympic Dam facility has been through several expansion phases, the latest of which employs Citect's SCADA software to replacing an aging DCS.

Expanded Mining Operations

While underground mining operations remain mostly unaltered, an extensive new plant maximizes the mine's ore extraction capabilities. The metallurgical plant features a new ore stockpile, a new autogenous mill, and additional flotation cells in the copper concentrator section. The hydrometallurgical plant expansion includes the tailings leach area and the copper-uranium solvent extraction area. A new smelter complex has a nominal capacity of 180,000 tpa copper, while an associated acid plant has a capacity of 1,640 tpd sulfuric acid. The copper refinery has been expanded to a nominal capacity of 179,000 tpa electro refined copper and 23,750 tpa electrowon copper. In addition, a new gold and silver refinery has been constructed.

OEP Establishes Requirements

To manage the expansion, WMC appointed an Olympic Expansion Project (OEP) team made up of WMC consultants and contractors.

The OEP team sought an open solution to their control system needs. The original facilities were controlled by an ABB Mod300 DCS system using proprietary hardware and software. The new facilities will have more than 400,000 real-time PLC variables, 40,000 physical I/O points and 20,000 serial I/O points. The new controls will use an open, non-proprietary system that can be easily maintained and expanded in the future.

In addition to the technical requirements, OEP wanted a partnership approach to the management and development of the system. Citect and Allen-Bradley offered OEP a combined solution using Allen-Bradley plant floor controllers (PLCs), the Citect SCADA package and a Microsoft SQL server. Ethernet was selected as the best communication method as the open, non-proprietary platform allowed a multitude of vendor hardware to be supported across the site. This new control system was chosen following a comprehensive analysis based on the technical solution offered, cost effectiveness and Citect and Allen-Bradley's partnering abilities.

Citect engineered the project over 18 months, with a manning level of 42 engineers at its peak. Project teams corresponded to designated process areas including: smelter, hydromet, concentrator, refinery, mine, auxiliaries, infrastructure and existing plant upgrade. A separate team addressed the development of project standards and documentation pertaining to the control system.

Development tools provided by Citect allowed a single development database for SCADA configuration, PLC programming, and much of the documentation, which greatly minimizing the engineering effort.

Information Anywhere

The SCADA software made it possible to view any information at any location on the network. While viewing facilities is global, control is limited by password security such that any station, with the appropriate password, can control distinct plant sections. If required, the entire plant can be monitored and controlled from a single SCADA station.

FIGURE 2.

CONTROL ROOM



The complete system hosts more than 148 PLCs and monitors 3,500 analog trends every two seconds.

Standards-based Solution

OEP and the software vendor applied rigorous standards for SCADA software, the SQL server and the PLCs. From tagging structures and graphical displays to program structures and documentation, the standards put in place ensured that, in addition to a consistent operator interface, the underlying PLC program and architecture were consistent plant wide, including vendor-supplied PLC systems.

The early development and documentation of standards allowed multiple engineers to work on parallel paths during the development phase. Furthermore, advanced generation tools could be used to auto-generate large portions of the code for both the SCADA software and PLCs. This greatly reduced the engineering time required for development and allowed the aggressive schedule to be met.

Using these standards reduced maintenance, training and development costs, and assured consistent results across the site.

The SQL servers retrieve historical trend data, alarm data, operator actions and system events from the SCADA system. In the event of failure, data is maintained on the SCADA system until the SQL servers have succeeded in retrieving it. In the event that a primary source of data is unavailable, the SQL servers retrieve information from the standby Citect SCADA server.

Performance was a critical factor in the design of this large PC-based SCADA system. Prior to delivery a complete soak test was performed to ensure that system performance would satisfy WMC's specification; OEP insisted on testing the system to 200% of its designed load.

The fully commissioned system will encompass 148 PLCs, the majority of which are Allen-Bradley Ethernet PLC5 series. The remainder consists mostly of Simatic PLCs and various other brands tied to the Ethernet. There will be 60 operator interface stations (Figure 2), 10 I/O Servers, two Trend Servers, two Alarm & Report Servers, two Windows File Servers and two SQL servers. The entire system runs on a single runtime database with access to all information from any operator station via security access.

Reliability and Integrity

Integrity and reliability of the system were of paramount importance to OEP. Redundancy was used in the following applications:

- Dual file servers provide primary and backup domain controllers if a file access error occurs.
- Dual SQL servers ensure complete data integrity on the Data Management System by using data replication. Furthermore, the SCADA software stores all data to be logged to the SQL servers locally prior to transfer such that in the event of SQL communications or server failures, data is not lost.



- Fully redundant 100-Mbps Ethernet switched network running TCP/IP over single- and multi-mode fiber optic cable is used between the control room (communications center) and distributed remote locations.
- Fully redundant 10-Mpbs unshielded twisted pair is used at each remote location and inside the control room itself, with 100-Mpbs switched links direct to all SCADA Servers.
- All remote computers and other selected computers have redundant network interfaces in each PC with one on each LAN, forming a completely redundant communication path to each of them.
- Most of the plant floor controllers are also Ethernet-based with dual Ethernet connections. The SCADA I/O servers automatically switch to the standby Ethernet connection on the PLC if the primary connection fails.
- The SCADA system provides for fully redundant hot fail-over server facilities for data acquisition and distribution (I/O server) functionality.
- The SCADA software provides for fully redundant hot fail-over server facilities for Trend, Alarm and Report servers. Each of these tasks is allocated to a specific pair of primary and standby servers.

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Inherent Performance

The SCADA system's built-in data optimization for communications to the plant floor controllers allows the large-scale application to be manufactured similarly to a small-scale application. Multiple I/O servers were used to localize the process areas.

The SCADA system automatically caches the data for each plant floor controller, thereby minimizing the communications required to distribute the data to multiple SCADA clients. This built-in functionality results in minimal performance impact on the entire system when large numbers of SCADA clients are added.

The SCADA software's scalability and data management allows it to provide extremely high performance on large-scale systems without increasing engineering complexity.

Editors note: *See controlglobal.com for expanded coverage.*

Digital Data Recorders Deliver

Mechanical chart recorders "need constant manual calibration and have moving parts which introduce errors due to friction and repetitive strain," says Gricha Raether, the industrial control and distributed I/O product manager at National Instruments (www.ni.com).

"Paper data can fade with time, and data storage is cumbersome because it involves storing thousands of sheets of paper in folders and warehouses. Data retrieval means going through all those sheets to find the correct time segment," concludes Raether.



"Local data logging can be the easiest way to satisfy regulatory requirements because data is stored before transmission and before human interface or intervention."

Analog chart recorders are inferior in other areas. "Chart recorders are limited to the process they are directly connected to, offer little in the way of higher level analysis, and cannot address ad-hoc analysis," says Roy Kok, the HMI/SCADA manager at GE Fanuc Automation (www.gefanuc.com).

"Digital recorders have reduced maintenance costs because of the elimination of mechanical parts such as chart drives and pen mechanisms," according to Douglas Bradway, the recorder product manager for Honeywell International (www.honeywell.com). "The elimination of consumables such as the charts, pens and ink wheels provides additional savings," adds Bradway.

Two innovations in data acquisition are of note. The first is the ability to digitally record and store data locally. The second is the ability to transmit this recorded data in industry standard file formats.

Recording data locally is simple, reliable and tamper proof. "We have implemented an on-board data logging engine inside our net concentrator I/O system," reports Scott Saunders, the director of strategic marketing for Moore Industries (www.miinet.com). "Our net concentrator system allows the user to pick and choose which I/O points are recorded locally, and to specify when to turn the data logger on and off."

Local data logging can be the easiest way to satisfy regulatory requirements because data is stored before transmission and before human interface or intervention.

Many of these local data recorders also provide data processing. "Our data monitors record process data for later retrieval," says Charlie Strack, marketing manager for STI Automation Products Group (www.stiapg.com). "Ethernet or a cell phone modem can be used to transfer the data to a host device. This technology offers high resolution, 24-bit data input channels with digital software-based signal processing capability," adds Strack.

Most local data storage devices comply with open standards such as OPC, XML, SOAP and .NET. Stored data can be transmitted via hard-wired or wireless means.

Stand-alone digital recorders are simple to use and work well for operators used to traditional chart recorders. "Our paperless recorders incorporate graphic displays to show trends, bar graphs, digital values or combinations of these basic displays. They can also provide customized displays, store electronic data for local recall, or enable data acquisition over a network," says Bradway of Honeywell.

Web-based data recorders serve data to the web for access via web browsers. "Our DataNet OPC software is a web-based presentation and logging software solution," reports William Glover, the product development manager at BizWareDirect (www.bizwaredirect.com). "DataNet uses OPC technology to display real-time data from industrial devices on a live web page," concludes Glover.

PC-based data recorders are part and parcel of most Windows-based HMI software. "Information once recorded on paper chart recorders is now primarily gathered, presented and archived electronically via PC-based solutions," says Steve Ludwig, the manager of commercial programs in the automation control and information Group at Rockwell Automation (www.rockwell.com).

"Depending on the informational requirement, plant floor data may be presented and archived by HMI software for operator use, middleware systems for management use, or moved into business systems for tracking and archiving," adds Ludwig.

Lower cost data recording can be implemented via CE-based devices. "With Historian Collector functionality added to a CE-based operator interface, a user can record data for a fraction of the cost of a chart recorder; under $800," according to Paul Daugherty, the manager of operator interfaces at GE Fanuc Automation (www.gefanuc.com).

Dan Hebert, PE, Senior Technical Editor
dhebert@putman.net

So Much To Learn, So Little Time

"Ask the Expert," on ControlGlobal.com, is moderated by noted process control authority Béla Lipták. Béla and his cadre of leading experts in process automation recruited from among the co-authors of the *Instrument Engineer's Handbook* 4th Edition, are "in the box" all month answering process control questions from all comers. The best will be published here.

Question

I am an operations engineer at a small oil blending facility. The plant is 50 years old and most of the equipment is at least that old. I have slowly been upgrading, mostly using high efficiency motors and Variable Frequency Drives (VFDs). Now comes the hard part. I need to learn as much as I can on control processes from the ground up. My first attempt has been to take the analog pressure gauges out of the tank farm and use pressure transducers with digital readouts with a 4-20 mA output. I have a lot to learn in a short time. Could you kindly recommend books, online courses etc. Any assistance would be kindly appreciated.

Answers:

My former controls professor, Dr. Carlos Smith, does excellent practical on-site courses for process engineers, new controls engineers, instrumentation technicians, etc., on process control. He is now currently Associate Dean of Academics at the University of South Florida's College of Engineering. Dr. Smith offers three popular courses: Automatic Process Control I, Automatic Process Control II, and Chemical Engineering for non-Chemical Engineers. Each can run 3-5 days on site, and he will adjust the content and level of the course to suit the audience. He does an excellent job of boiling down control theory to its most essential and practical parts and conveying it in terms anyone can understand and apply. He has conducted these courses for Monsanto, Dow, Procter & Gamble, Weyerhaeuser, and many more. Contact Carlos Smith, Ph.D., Associate Dean of Academics, University of South Florida, Tampa, Fla.; 813/974-5859; csmith@eng.usf.edu.

And if an on-site course is not an option, then I also recommend the following books:

- "The Instrument Engineer's Handbook" Series, Bela Liptak, CRC Press
- "Principles and Practice of Automatic Process Control," A. Corripio and C. Smith, Wiley.
- "Feedback Controllers for the Process Industries," Shinskey, McGraw Hill.

Rick Meeker, Jr., P.E.,
Process Control Solutions, Inc.

Try OSU OK?

For those with a BS degree in an engineering field, Oklahoma State University offers a distance education Masters of Science in Control Systems Engineering. It is a 33-SCH (semester credit hour) (11 courses) graduate engineering program, requiring a 3.25 or better undergraduate GPA for admission, and is aimed at the practicing engineer who wants preparation for industrial control. Courses are interdisciplinary and taught by chemical, electrical, mechanical, and industrial engineering professors. Visit www.okstate.edu/ceat/mscse for information.

R. Russell Rhinehart,
Bartlett Chair and School Head,
School of Chemical Engineering

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ISA's distance learning programs can be found online at www.isa.org.

Dale Lee
Director
Convention, Education, and Certification Services, ISA

NEED AN EXPERT OPINION?

Send your knotty control and automation questions to Béla Lipták at bliptak@putman.net. Lipták will either answer the questions himself or bring in specialists from his cadre of *Instrument Engineer's Handbook*, Fourth Edition co-authors. Questions and answers will be published online periodically on ControlGlobal.com as they are received.

Each month, Béla will pick the best question, and publish it, along with the answer in CONTROL. If your question is chosen, a prize (of the editorial staff's choosing) will be awarded to the submitting winner.

Think About BCIT

I think British Columbia Institute of Technology (BCIT) has very good courses in the area mentioned by the reader. The courses range from Tradesman's tickets in Instrumentation to Bachelor Degrees in Controls Engineering. Of course I might be a bit biased, but the fact that BCIT has won the ISA student competitions repeatedly over the past five years is a good indicator. Finally, the facilities are first class for a practical education in controls–it was one of the reasons I decided to come here to do my research work.

Eric Byres, PE,
Research Manager
BCIT Internet Engineering Lab

Meet Me in St. Louis

Washington University in St. Louis has an excellent program for in Chemical Engineering particularly for people who want to work in the area of process control because it uses a virtual plant that consists of a series of industrial advanced control tools and high-fidelity dynamic simulations and ISA books that help prepare the individual for the CAP program.

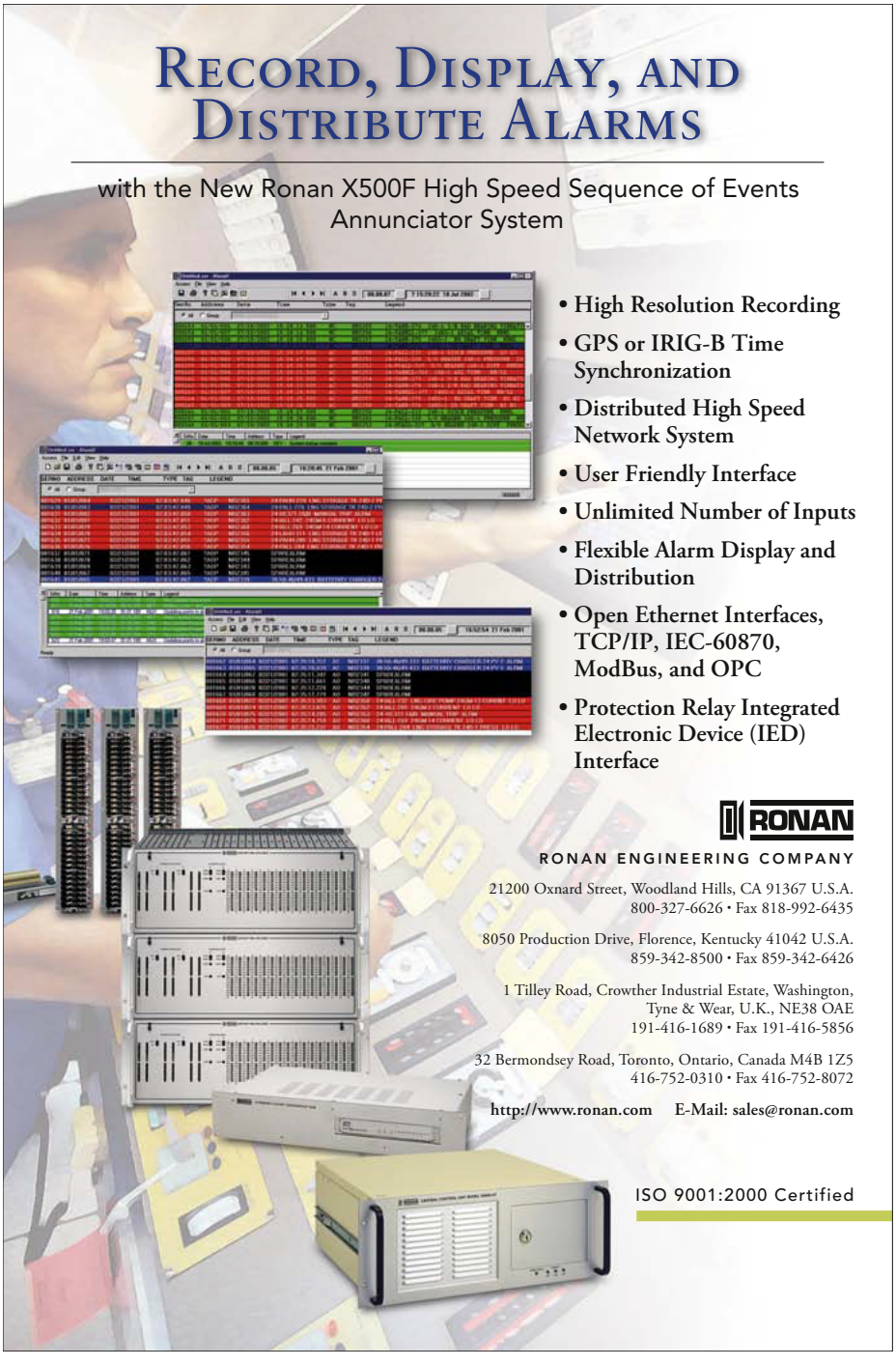
Greg McMillan
Control Columnist

Try Your Local Chapter

I would suggest:

- Join your local ISA chapter. You'll probably meet a lot of guys who can tell you all about it, as well as what new stuff might work well in it's place.
- Run out and buy a copy of the "Instrument Engineer's Handbook."
- Contact Omega at www.omega.com and ask for a complete set of their catalogs. At the back of each section of the catalogs is a very well-written, short explanation of each type of measurement device.
- Pick up a copy of "Fundamentals of Process Control Theory" by Paul W. Murrill, Ph.D. ISA press ISBN: 0-87664-507-4.
- And don't forget "Measurement and Control Basics" by Thomas A. Hughes, ISA press.
- In addition, the ISA has lots of good courses on these topics.

David Gardellin
Onyx Valve Co.



Process Analyzers Avoid Meeting Standards

NeSSI and PAT Standards are Being Ignored by Most Analyzer Manufacturers

The newest standards in the process analyzer world are NeSSI and PAT. The NeSSI Initiative/ANSI/ISA76-99.02 (2002) standard defines modular sampling systems that make it extremely easy to set up the manifolds, valves, tubing, piping and sample conditioning systems needed to bring process samples to analyzers. It also defines a 1.5 x 3 x 3-in. format for plugging analyzers onto a NeSSI board.

The Process Analytical Technology (PAT) Initiative recommends procedures for improving manufacturing of food and drugs through measurements of materials and processes. It uses NeSSI and ISA-76 to help define procedures that will simplify measurements and reduce the costs of analyzers and components.

Except for one single product in the roundup that follows, manufacturers seem to be avoiding these standards and recommendations like the plague. NeSSI components cost a bit more than standard sampling systems, so we suspect manufacturers are trying to avoid increasing prices in these difficult economic times. Instead of developing products that might not sell because of price, they spend what little R&D monies available on enhancements, ports, software and interfaces.

Robert Sherman, industry specialist at CIRCOR Instrumentation Technologies (www.circortech.com) says that such instrumentation is on the way. "Analysis elements currently commercially available on the NeSSI format are limited to ppm oxygen and ppm moisture," he says (www.spectroscopyonline.com). "As soon as volume increases, one of the NIR or Raman probe manufacturers will machine a probe in this format. Complete spectrophotometers and gas chromatographs currently are under active development."

Even if they are ignoring the standard, the enhancements to standard analyzers are downright impressive. Analyzers are getting easier to configure and connect to control systems, they operate much closer to real-time than ever before, and they are getting more rugged so you can install them closer to the process.

Standard features often include Windows-based HMI, configuration and data analysis software, color displays, touch screens, built-in data loggers, and standard interfaces such as wireless, 4–20 mA, USB, and infrared to PDAs and Pocket PCs. Nary an IEEE-488 interface–the old instrumentation standard—can be found these days in new analyzers.

For the present time, process analyzers appear to be immune to the "China Syndrome" that afflicts so many other control devices: i.e., companies in Asia copying designs, and then making and selling products at extremely low prices. Process analyzers are bit too complex to be stolen … er, emulated in such a manner, so North American, Japanese and European vendors continue to dominate the market.

The analyzers we list in the round-up that follows have so many new features and functions, you may want to go to ControlGlobal.com to view all the product information we have on several of them. Curious? Go to ControlGlobal.com and look for the round-up section under "Site Highlights."

–Rich Merritt, Senior Technical Editor

MOISTURE ANALYZER MONITORS ITSELF

The rack-mounted Model 3050-RM Moisture Analyzer has an internal moisture generator allows users to confirm analyzer performance on schedule or on demand. The on-line verification system verifies the zero point of the sensor's calibration by comparing sensor and system operation to a conditioned sample gas that contains a known amount of moisture. The analyzer is suitable for applications that require continuous measurement of ppmv moisture in gases and vapor. More info at controlglobal.com. **Ametek Process Instruments; 800/222-6789; www.ametekpi.com**



WIRELESS PH TRANSMITTER

The OM-CP-RFPH101A measures pH and temperature, transmits readings back to a host computer via wireless, and logs readings in its memory. A slide switch allows the transmitter to be turned on or off without affecting operation of the device. Windows software makes a PC into a real-time strip chart recorder, prints data in graphical or tabular format, and exports data to a text or Microsoft Excel file. Price: $449. **Omega Engineering: 203/359-1660; www.omega.com**



DUAL ANALYZER MULTI TALENTED

The UDA2182 Universal Dual Analyzer monitors and controls pH,

Process Analyzers

ORP, conductivity/resistivity, concentration or dissolved oxygen. It has a graphic backlit LCD display providing continuous readout of process variables, alarm and diagnostic status, infrared communications for configuration via a Pocket PC, laptop or desktop computer using Process



Instrument Explorer software. The analyzer has dual 4–20 mA current outputs and dual relay outputs for control. More info at controlglobal.com. **Honeywell; 215/641-3798; www.acs.honeywell.com**

HUMIDITY TRANSMITTER WARMS TO THE TASK

The HMT330 Humicap Series includes six models covering humidi-



ty measurements from 0–100 %RH, temperature ranges between -70° and +180° C, and pressure ranges between 0 and 100 bar, depending on the model. In very humid or chemically harsh conditions, a warmed probe and chemical purge help maintain measurement reliability. Outputs include a numerical and graphical display with multilingual user interface, three analog outputs, serial interfaces and alarm relays. **Vaisala; 781/933-4500 X1063; www.vaisala.com**

GAS CHROMATOGRAPH PLAYS A NEW STANDARD

An ASTM designation has been assigned to the PGC2007 Gas Chromatograph: ASTM D7041-04, "Standard Test Method for Determination of Total Sulfur in Light Hydrocarbons, Motor Fuels, and Oils by Online Gas Chromatography with Flame Photometric Detection." The test method covers the determination of total sulfur in liquid hydrocarbons with



a final boiling point less than 450° C by gas chromatography using a flame photometric detector. More info at controlglobal.com. **ABB; 304/647-1761; www.us.abb.com**

SPECTROMETER GOES LOW

The QE65000 Spectrometer can achieve up to 90% quantum efficiency with high signal-to-noise and rapid signal processing, making it suitable for



low-light level applications such as fluorescence, DNA sequencing, astronomy and Raman spectroscopy. It has a Hamamatsu back-thinned detector with

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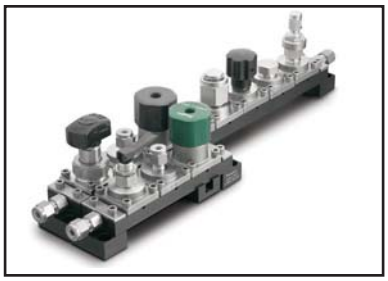
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Process Analyzers

a 2-D arrangement of pixels (1,044 horizontal x 64 vertical) that is responsive from 200–1100 nm. Price is $9,999. More info at controlglobal.com. **Ocean Optics; 727/733-2447; www0ceanOptics.com**

SAMPLE HANDLING GOES MODULAR

Modular platform components and configurator software make it easier to develop and assemble ANSI/ISA 76.00.02-



compliant process analyzer and sample-handling systems. Modular platform components consist of surface mounts, including shut-off, needle, metering, toggle and check valves, as well as filters, substrates and manifolds. The configurator identifies all the flow connectors that are necessary to build the fluid system and generates a bill of materials and assembly diagram. **Swagelok; 440/349-5934; www.swagelok.com**

BETTER THAN AVERAGE OXYGEN ANALYZER

The O2mation model AV550 multichannel averaging oxygen analyzer has a 5.7-



in. color touch screen, trend graph functions and process diagnostic tools. The display can show shows average data, single detectors or all detector gas concentrations. The analyzer can accept inputs from eight oxygen detectors, making it suitable for power boiler oxygen trim and combustion control in large utility boilers or various industrial furnaces. **Yokogawa; 770/254-0400; www.yokogawa.com**

PROBES PLUG-AND-PLAY

Sensors for pH, ORP, conductivity, dissolved oxygen and turbidity are plug-and-play with the SC100 Digital Controller; that is, the controller accepts any two sensors without special cabling, configuration or software. The



controller has a built-in data logger that stores readings from the sensors for up to six months, along with calibration and verification data points, alarm history, and instrument setup changes in CSV format. **Hach; 800/227-4224; www.hach.com**

OXYGEN ANALYZER HANDLES HIGH TEMPS

The Oxymitter oxygen transmitter operates in temperatures up to 1,300° F, making it suitable for making in-situ meas-



urements during the annealing of cold rolled metal products. It measures oxygen levels from 0.05–0.1% with accura-

Process Analyzers

cy of 0.05%. Because it can be inserted directly into a furnace, the device needs no sampling system. **Emerson/Rosemount Analytical; 330/684-4579; www.emersonprocess.com**

SULFUR ANALYZER RESPONDS QUICKLY



The SOLA II Sulfur Analyzer responds to a change in total sulfur concentration immediately following sample injection, typically every 30 seconds, providing the user a rapid indication of the rate of total sulfur concentration change. Using pulsed ultraviolet fluorescence spectrometry (PUVF), it delivers total sulfur measurements from 2 ppm to 25 ppb in liquid phase petroleum fractions with end boiling points less than 400° C. **Thermo Electron; 713/272-4516; www.thermo.com**

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Process Analyzers

DIGITAL PH SENSOR

The Memosens digital pH sensors have digital memory functions that save measuring point data, the site, operating hours, calibration data, and minimum and maximum pH values. The sensor can be precalibrated in



the lab under optimum conditions and then simply replaced at the measuring point. **Endress+Hauser; 888/end-dress; www.us.endress.com**

ORP AND PH SENSOR DIFFERENTIATES

WDS Series sensors have a differential measurement technique that uses two electrodes, one for the process measurement and the other for the reference measurement. Each electrode is measured differentially with respect to a



third metal electrode. If the glass reference solution becomes contaminated, or the salt bridge becomes clogged, it is easily replaced. The glass electrode is not prone to chemical attack. **Walchem; 508-429-1110; www.walchem.com**

CONDUCTIVITY PROBE TESTS ULTRA PURE WATER

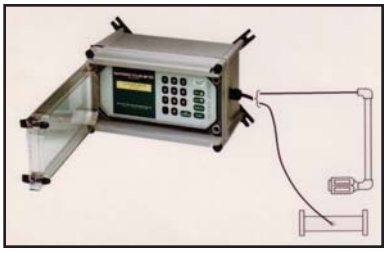
The InLab 740 compensates for the change in conductivity in a solution as temperature increases or decreases through use of an integrated temperature sensor, making it suitable for lab use. It



ships with a test certificate stating the actual cell constant of each individual sensor. **Mettler Toledo; 614/438-4853; www.mt.com**

SUSPENDED SOLIDS METER

Series 502 uses ultrasonic attenuation to measure the concentration of sludges and slurries that are too thick for optical methods. It has automatic pushbutton



calibration, ultrasonic self cleaning, self diagnostics, error-trapping algorithm, automatic dynamic damping, and a 4–20 mA output that is linear with percent suspended solids. It is available with two styles of pickup heads. The instrument is packaged in a NEMA-4X enclosure with a weatherproof clear cover. **Markland Specialty Engineering; 416/244-4980; www.sludgecontrols.com**

PIPELINE ANALYZER

Detcon Model series 1000 process analyzer provides continuous, real-time measurement of H_2S and CO_2 concentrations in natural gas pipelines, tail gas lines and other gas systems. Available

Process Analyzers

ranges are from 0–20 ppm up to 0–10,000 ppm for H2S, and from 0–3,000 ppm up to 0–100% for CO2.



The instrument is rated for use in Class I, Division 1 and Group C and D areas. Line, battery and solar- powered models are available. **Detcon, Inc.; 281/367-4100; www.detcon.com**

MOUNTS DIRECTLY

MicroSAM on-line process gas chromatograph (GC) uses a thermal conductivity detector array integrated on silicon to measure all column and vent flows as well as the injection peak itself.



It has valveless live injection, software-adjustable injection volume, maintenance-free column switching, electronic pressure control and an explosion-proof enclosure. **Siemens Energy & Automation; 918/662-7343; www.siemens.com**

RACK-MOUNT FTNIR

The Analect Diamond MX FTNIR analyzer provides multiple sample process measurements of liquids, solids, pellets and powders. It uses fiber coupled sampling devices from an integrated mux, which switches the entire beam over nine channels. Sampling options include cells for transmission or diffuse reflectance. **Applied Instrument Technologies; 909/593-3581; www.hs-ait.com**



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CorrTran Corrosion Detection Transmitter detects and reports corrosion or pitting of pipes or vessels in Class I, Div. 2 hazardous locations. It has a two-wire,



loop powered 4–20mA signal with HART protocol. A variety of probe styles and materials of construction are available. **Pepperl+Fuchs; 330/486-0002; www.am.pepperl-fuchs.com**

ADJUSTABLE SPEED CONTROL

Model H2 drive is available in standard ratings of 1–450 hp, and in custom



models up to 1,500 hp. All models have a large graphical display, extensive diagnostic information, and interfaces to DeviceNet, Profibus, Lonworks and Ethernet. Setup is via the company's Workbench software from a PC, or via a server over the Web. **Baldor Electric; 479/646-4711; www.baldor.com**

SOFTWARE COMPLIES WITH FDA RECORD-KEEPING RULES

The eCompliance Solution package helps food companies meet the requirements of the reporting and record-keeping rule announced by the U.S. Food and Drug Administration on Dec. 9, 2004. The regulation requires companies to establish, maintain and, upon request, provide access to records as directed by Congress in the Public Health Security and Bioterrorism Preparedness and Response Act of 2002. More information on the act and the software at controlglobal.com **Wonderware; 949/753-9292, ext. 244; www.wonderware.com**

KEY OPERATED SAFETY INTERLOCK SWITCHES

GKN Series safety key operated switches have positive opening safety contacts and double insulation that meets IEC 60947-5-1 requirements. Options include choice of five switch contact configurations, including two configurations with three contacts, and three standard key actuators, including a flexible key actuator that allows for minor misalignment. The rotating head allows actuator engagement from the front, back or either of two top positions for a total of four different actuation entry points. **Honeywell; 800/784-3011; www.honeywell.com**

SENSORS SAFE FOR SEMICONDUCTORS

All exposed components of the QF5.5 capacitive level sensor are semiconductor process safe and will not harm or be harmed if they come into contact with



materials used in semiconductor manufacturing. It has a covered potentiometer and an all-plastic polypropylene housing for applications that do not permit metal in the process area. More info at controlglobal.com. **Turck; 800/544-7769; www.turck.com**

TEMPERATURE CONTROLLERS PLUG AND PLAY

Plus (+) Series DIN temperature controllers have plug-in modules for

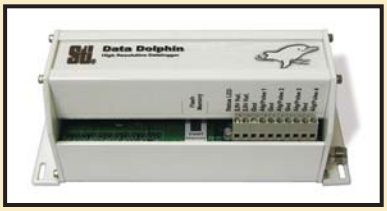


outputs, inputs, alarms and communications, so that users can keep only

one type of controller in stock and adapt it for each application as required. Models include 1/16, 1/8 and 1/4-DIN size units sold under West and Partlow brands. Controllers have dual setpoints that can be selected from a remote location, and PID control with pretuning and self tuning. More info at controlglobal.com. **Danaher Industrial Controls; 800/873-8731; www.dancon.com**

DATA MONITOR HANDLES TOUGH ENVIRONMENTS

The Data Dolphin 220 Data Monitor operates in temperatures from -40–60∞



C. It has a 24-bit A/D that accepts four differential bipolar inputs with resolution to 90 nV, which allows most sensors to connect directly without any signal conditioning. Four additional inputs allow the user to select pulse, status, event start/stop or analog capability. More info at controlglobal.com. **Scientific Technologies; 888/349-7098; 435/753-7300; www.stiapg.com**

CONTROLNET CARDS PLUG AND PLAY

Models 1784-PCIC and 1784 PCICS plug-and-play cards can detect and adapt to varying voltage amounts, so users can apply them to virtually any industrial or personal computer using a PCI bus and link their applications to any ControlNet network. Both cards can work with Rockwell Software RSLinx server software, and the 1784-PCICS card can work with SoftLogix PC-based control software. Drivers also are available for Windows NT/2000/XP operating systems. **Rockwell Automation; 800/223-5354, Ext. 1778; www.ra.rockwell.com**

EXPLOSION VENT PROTECTS PANELS

ATEX-approved Panel Flo-Tel burst-detection device works with the company's explosion vent pressure-relief



devices to provide a signal once a panel has operated. The system provides simple, fail-safe detection to minimize the risk of spurious alarms. The explosion vents operate over a pressure range of 0.7– 7 psig and in temperatures from -40–400° F. **Elfab; 877/882 2880; www.elfab.com**

TAKES LIGHTNING STRIKES

MA3100 Series surge protectors are available in three models to meet IEC 61312 Class I, II and III specs for a lightning current arrester, surge current arrester, and surge protector, respectively. The Class I device is rated for 60kA, fits on a DIN rail, and is 18-mm wide. Each device connects in parallel with the power system via a fused spur. More specs at controlglobal.com. **MTL Surge Technologies; 800/544-4737; www.mtlsurge.com**

EMBEDDED DAQ SYSTEM

Model E4-DAS PC/104-based data acquisition system is packaged in a lightweight aluminum enclosure with a railed card-cage subassembly and internal elastomer inserts, making it suitable for embedded applications requiring protection from vibration and shock. It has a low-power 133-MHz Pentium PC/104 CPU and peripheral interfaces including VGA, keyboard and mouse, 10/100BaseT Ethernet, and quad RS-232 ports. Price: $1,195.00 More info at controlglobal.com. **ACCES I/O Products; 858/550-9559; www.accesio.com**

WIRELESS SYSTEM MONITORS PLANTS

Skymetry monitors fluid storage, flow meters, water levels, timers, temperature, power failure and alarms at industrial and chemical plants. When hazardous conditions occur, it alerts facility managers via wireless voice messages or email. The system monitors and controls equipment and environmental conditions using eight dry inputs, six analog inputs, built-in power failure detection and two relay outputs. More info at controlglobal.com. **Sensaphone; 877/373-2700; www.Sensaphone.com**

PRESSURE RELIEF VALVES



Type E thermoplastic pressure relief valve has a PTFE bellows stem seal and is available in body materials of PVC, PP, PVDF and PTFE, in sizes from 1/2 to 4 in. O-Ring seals are EPDM, FKM or PTFE-encapsulated FKM. The pressure-relief valves can be installed in thermoplastic piping and metal pipelines, are corrosion resistant, and can be used in aggressive chemical applications. **Asahi/America; 781/321-5409; www.asahi-america.com**

RING-A-DING SWITCH

The ET-5RS EtherTrak real-time ring switch manages Ethernet networks, provides traffic prioritization and fault-tolerance, and works in industrial environments. Priority assignments can be set by the user, recovery time is less than 5 msec per hop, and it allows 50 switches to be connected in a ring. Options include copper and fiber ports for connections up to 40 km. The switches can be pre-configured or field configured with a software wizard. **Sixnet; 518/877-5173; www.sixnetio.com**

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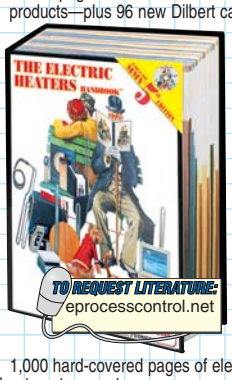
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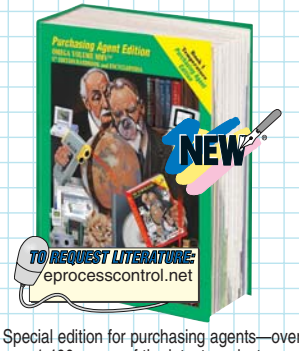
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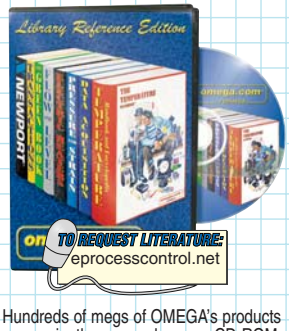
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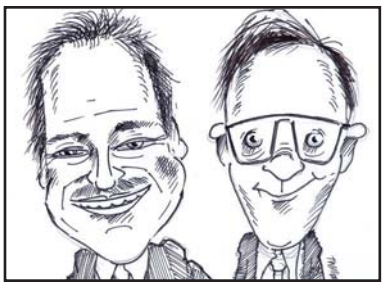
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Peak Concerns



Greg McMillan (gkmcmi@msn.com) and Stan Weiner, PE (sweiner@swfla.rr.com) bring their wits and more than 66 years of process control experience to bear on your questions, comments, and problems.

Stan: The interest in the February puzzler peaked with some early replies from Bob Cameron and Glenn Mertz who correctly surmised why a batch temperature controller with a single steam valve overshot its setpoint in the heat-up phase.

Bob: The controller had too much reset action.

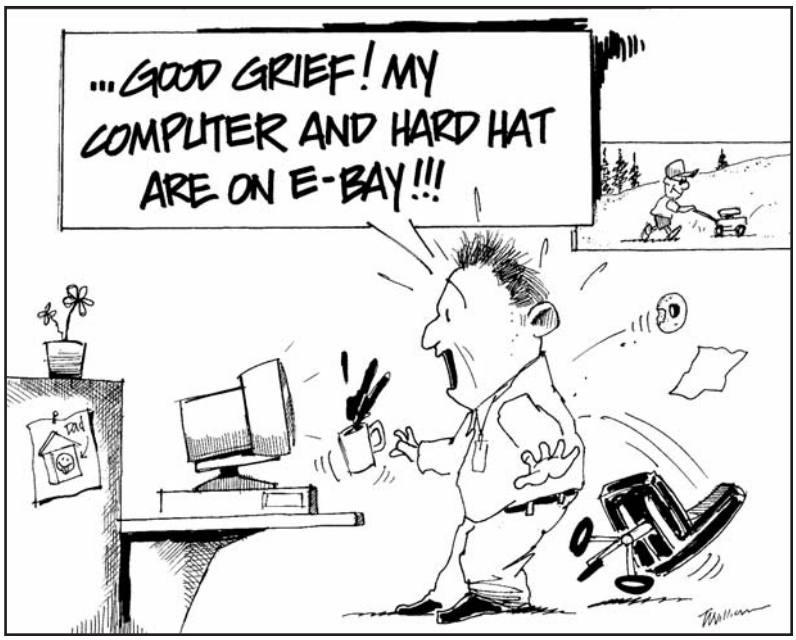
Glenn: The controller also didn't have enough rate action.

Greg: Reset action doesn't cut back on the heat until the temperature crosses the setpoint, which is too late especially since there is no cooling. The temperature will continue to coast after the steam valve closes and reach a peak after the transpiration of the system dead time. Rate will provide some anticipatory action by acting on the rate of change of temperature.

Stan: The total system dead time can be approximated as the sum of the process dead time, thermowell lag time, valve dead time, and controller execution time. Some batch control systems calculate the rate of change of the temperature and multiply it by the system dead time to get an incremental temperature that is added to the present temperature to get a projected temperature. When the projected temperature exceeds its setpoint, the steam valve closes. The rate of change calculation is done per our March column ("Time after Time," p70) to maximize the signal to noise ratio and minimize any extra dead time.

Greg: What is really needed is a proportional plus derivative (PD) controller to prevent overshoot for this zero-load system. However, this is a change in structure that may require validation. At a very minimum, a simulation is needed to test the controller's response to changes in its mode, output, or setpoint by the batch sequence. In particular, the bias of a PD controller must be set properly and retain its value for different modes and tracking options. Batch operations tend to exercise many modes. It is common for a batch sequence to switch the controller among the manual, automatic, remote output, and remote cascade modes. For this zero-load application, the PD bias should ensure the controller output is zero at a temperature sufficiently below the setpoint so that incremental change in temperature during the system dead time doesn't cause overshoot. The controller output can also be switched when the temperature approaches setpoint from throttling to modulation of a pulse width where the pulse cycle time is the system dead time.

Stan: Other examples of zero-load applications are the addition of a nonreactant to a batch by concentration control, the filling of a batch by level control, and the neutralization of a batch by pH control. For neutralization, acid and base valves are sometimes used to recover from overshoot, but this leads to the split ranging issues noted in the February column ("Split Personality," p83) and a loss in batch cycle time and efficiency.



Greg: Much of the benefit of the PD controller can be achieved in a PID controller by simply increasing the reset time by two or more orders of magnitude and increasing the rate time to be about the total system dead time. The controller gain also often needs to be increased as well by 50% or more. A controller with negligible reset action does not drive the process variable (PV) to the setpoint or center the proportional band window on the setpoint (SP). The user must take care of two things to pull off the startup or batch sequencing of this loop. First, the controller output must be initialized to a proper value before being switched to automatic, cascade or remote cascade. Second, the SP must track the PV in manual, remote output and local override modes, and the SP must be moved to the desired end point or operating point after the controller is switched to a mode such as cascade or remote cascade. Otherwise, the proportional band window is located at the present PV, which causes premature shutoff of the valve in automatic or remote cascade even though a sizable valve position was initialized. I have heard very experienced engineers claim something was wrong with the PID algorithm for filling up an evaporator on level control for startup because the feed valve closed as soon as the level increased from zero. The level never got close to its constant SP even if the valve was initially wide open because the controller had a high gain setting and large integral time setting and SP tracking of PV was not enabled. The location and movement of the proportional band window is rarely discussed ever since controllers changed from proportional band to gain settings for the proportional mode.

Stan: Besides digging out industrial technical guides from the 60s and 70s, the user should use simulations to fully explore these and other confusing scenarios before commissioning a loop.

Greg: I would love the opportunity to see what a spin-off company could do with a zero load. I would be willing to help provide enough proportional and rate action for capital projects. Unfortunately, the parent company sees a spin off as an opportunity to dump its debt. Bob Cameron, Glenn Mertz and Mark Sowell offer the Top 20 signs a spin off may be spinning towards bankruptcy.

Top 20 Signs a Spin Off May Be Spinning Towards Bankruptcy

20. As a parting gift your new company gets 35% of the assets and 90% of the debt.
19. You stop receiving free copies of the trade journals.
18. The bathrooms suddenly have toilet paper dispensers that accept quarters.
17. The "prior convictions" question is omitted from the company's employment application.
16. Human Resources laugh when you ask to sign up for the stock-based 401K plan.
15. There's a toll booth at the entrance to the company parking lot.
14. You see an ad in the Sunday paper for a real estate open house at your company's address.
13. There have been an unusually high number of visitors, all with tape measures and log books.
12. You find paper airplanes around the executive offices made from company stock certificates.
11. The only cost-reduction plan implemented is head count reduction.
10. The CEO gets a huge bonus for losing less money than the budgeted loss. (Oh heck, give him one even if it's more, just to ensure that the captain goes down with the ship).
9. The new medical benefits plan consists of a pair of dice.
8. You are issued Monopoly money to pay your suppliers.
7. The executives are replaced with lawyers.
6. Domino's won't accept your check.
5. You have three phone messages from your financial planner.
4. You see your boss at the mall buying a new "interview" suit.
3. The evening shift Wal-Mart Greeter vacancy was filled by one of your colleagues.
2. There are orange survey flags along the plant property lines.
1. The janitor is your DCS supplier's representative at your company's Vendor Alliance meeting.

This Month's Puzzler: Why Did the Vortex Meter Go Bonkers?

Why did the vortex meter go bonkers when used to measure the flow of toothpaste?

Send an e-mail with your answer to the Puzzler, CONTROL questions, or comments to controltalk@putman.net.

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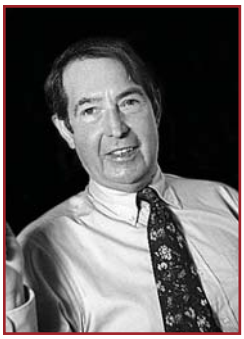
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Three Decades of DCS Technology

This year, Distributed Control System (DCS) technology marks 30 years since the first TDC-2000 beta-types were installed at Exxon's Sarnia, Ontario refinery, and other sites. During this period, the DCS has swept alternative process control technologies from the field.

At least two generations of process control engineers have matured during the DCS era, which has endured much longer than earlier technologies. The pneumatic and electronic-analog eras lasted 15–20 years each while computer-based control systems, overlaying various primary control eras, spanned perhaps two decades from the late 1950s to the late 1970s.



"DCS technology has had a profound effect on our profession. The pioneers deserve recognition for this substantial contribution."

DCS technology derives from direct digital control (DDC), the theory of which germinated very early in the digital epoch. The ENIAC (electronic numerical integrator and computer) machine at the University of Pennsylvania in 1946–1949 was the beginning of the stored-program, digital computer era. In 1952, Long and Holtzman of Shell Development discussed the feasibility of DDC in the Transactions of ASME. Monsanto Chemical first reduced DDC to practice in an ammonia plant in Luling, La., in 1962. Computer control pioneer Ted Williams, an inductee of the CONTROL Process Automation Hall of Fame, was the intellectual driving force behind this installation.

Honeywell had been working on a dedicated digital loop controller since 1969 but the problems of reliability and cost seemed insurmountable. The Intel 8080, announced in early 1974, was the breakthrough event enabling multiple control loops to be handled by a single microprocessor.

Honeywell's focus on the digital loop controller was motivated by their need to leapfrog the competition. Foxboro's SPEC 200, announced in 1972, was the high-water mark of analog electronic technology and ISA's 4–20mA standard (SP50) was approved in 1975 just as this technology was being rendered obsolete by digital controllers.

The TDC-2000 was an immediate success in the marketplace. Bailey Controls, under the strategic guidance of Bud Keyes, introduced the first significant competitive product (NETWORK 90) a year or two later. Fisher Controls and Fischer & Porter soon followed. Foxboro and Taylor Instruments, hoping to preserve their analog franchises as long as possible, joined this trend in the early 1980s.

Distributed control systems, because of their ability to integrate large numbers of measurements into the process control strategy, significantly increased the demand for transmitters and other peripherals. Shipments by U.S. manufacturers surpassed the $1 billion mark in 1989. Total shipments of all process control instrument products were just over $1 billion in 1972 at the high point of the analog era.

DCS shipments stayed strong and stable through 2001. During 2002-03 they dropped precipitously reaching a level in 2003 not seen since 1986. It may be that the value-added from this activity is being reported as software or services in some other category. DCS technology has had a profound effect on our profession. The pioneers deserve recognition for this substantial contribution.

U.S. Manufacturers Shipments – Electronic Systems Non-Unified Architecture

YEAR	SHIPMENTS ($ in millions)	YEAR	SHIPMENTS ($ in millions)
2003	$689	1991	$954
2002	$835	1990	$1095
2001	$1400	1989	$1039
2000	$1492	1988	$860
1999	$1484	1987	$778
1998	$1225	1986	$690
1997	$1428	1985	$705
1996	$1546	1984	$652
1995	$1432	1983	$441
1994	$1352	1982	$519
1993	$1307	1981	$420
1992	$1236	(Classification did not exist prior to 1981)	

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